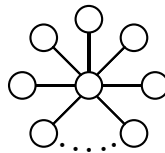


Examples:

1. As with the length of lists, it is relatively easy to have many 1s in a multiplicity list. The more interesting issue is how few of 1s may occur among lists for a given tree T . It certainly depends upon the tree, as the star (see Figure 34.8) may have just two 1s, while a path (see Figure 34.9) always has as many as the number of vertices.
2. If T has a diameter that is large relative to its number of vertices (a path is an extremal example), then it may have to have a minimum number of distinct eigenvalues, which forces $\mathcal{U}(T)$ to be much greater than 2. However, $\mathcal{U}(T)$ may be greater than 2 for other reasons. For example, for the tree T in Figure 34.10, $d(T) = 4$, $n = 8$, but $\mathcal{U}(T) = 3$. It is not known how $\mathcal{U}(T)$ is determined by T , and it appears to be quite subtle.

**FIGURE 34.8** A star.

34.5 Existence/Construction of Trees with Given Multiplicities

Definitions:

For a given graph G , the collection of all **multiplicity lists** is denoted by $\mathcal{L}(G)$. If it is not clear from the context, we will distinguish the unordered lists as $\mathcal{L}_u(G)$ from the ordered lists $\mathcal{L}_o(G)$.

General Inverse Eigenvalue Problem (GIEP) for $\mathcal{S}(T)$: Given a vertex v of a tree T , what are all the sequences of real numbers that may occur as eigenvalues of A and $A(v)$, as A runs over $\mathcal{S}(T)$?

Inverse Eigenvalue Problem (IEP) for $\mathcal{S}(T)$: What are all possible spectra that occur among matrices in $\mathcal{S}(T)$, T being a tree?

A tree T has **equivalence of the ordered multiplicity lists and the IEP** if a spectrum occurs for some matrix in $\mathcal{S}(T)$ whenever it is consistent with some list of ordered multiplicities of $\mathcal{L}_o(T)$.

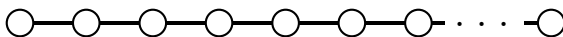
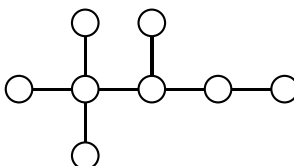
Facts:

1. [Lea89] Let T be a tree on n vertices and v be a vertex of T . Let $\lambda_1, \lambda_2, \dots, \lambda_n$ and $\mu_1, \mu_2, \dots, \mu_{n-1}$ be real numbers. If

$$\lambda_1 < \mu_1 < \lambda_2 < \dots < \mu_{n-1} < \lambda_n,$$

then there exists a matrix A in $\mathcal{S}(T)$ with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, and such that, $A(v)$ has eigenvalues $\mu_1, \mu_2, \dots, \mu_{n-1}$.

2. For any tree T on n vertices and any given sequence of n distinct real numbers, there exists a matrix

**FIGURE 34.9** A path.**FIGURE 34.10** A tree T on 8 vertices with $d(T) = 4$ and $\mathcal{U}(T) = 3$.

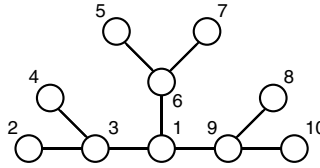


FIGURE 34.11 A tree for which there is not equivalence between ordered multiplicity lists and the IEP.

in $\mathcal{S}(T)$ having these numbers as eigenvalues.

3. Any path has equivalence of the ordered multiplicity lists and the IEP.
4. [BF04] There exists a tree for which the equivalence of the ordered multiplicity lists and the IEP is not verified. (See Example 1.)

Examples:

1. For remarkably many small trees and many of the families of trees to be discussed in the next section, the ordered multiplicity lists are equivalent to the IEP. This is not always the case. Extremal multiplicity lists for large numbers of vertices can force numerical relations upon the eigenvalues, as in the tree T shown in Figure 34.11.

Let $\mathcal{A}$ be the adjacency matrix of T (i.e., the 0,1-matrix in $\mathcal{S}(T)$ with all diagonal entries 0). The Parter–Wiener Theorem (Fact 5 Section 34.1) guarantees that $\alpha_{\mathcal{A}}(0) = 4$, and likewise guarantees two eigenvalues of multiplicity 2 (the nonzero eigenvalues of $\mathcal{A}[\{2, 3, 4\}] = \mathcal{A}[\{5, 6, 7\}] = \mathcal{A}[\{8, 9, 10\}]$, so the ordered multiplicity list of $\mathcal{A}$ is $(1, 2, 4, 2, 1)$. In fact, direct computation shows that $\sigma(\mathcal{A}) = (-\sqrt{5}, -\sqrt{2}, -\sqrt{2}, 0, 0, 0, 0, \sqrt{2}, \sqrt{2}, \sqrt{5})$.

However, it is not possible to prescribe arbitrary real numbers as the eigenvalues with this ordered multiplicity list. If $A = [a_{ij}] \in \mathcal{S}(T)$ has eigenvalues $\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4 < \lambda_5$ with multiplicities 1, 2, 4, 2, 1, respectively, then $\lambda_1 + \lambda_5 = \lambda_2 + \lambda_4$. The method used to establish this restriction comes from [BF04]. By the Parter–Wiener Theorem and an examination of subsets of vertices, $a_{11} = \lambda_3$. The restriction then follows from comparison of the traces of A and $A(1)$.

It is not known for which trees the determination of all possible ordered multiplicity lists is equivalent to the solution of the IEP. Even when the two are not equivalent for some ordered list, they may be for all other ordered lists.

2. In the construction of multiplicity lists for a tree, it is often useful (and, perhaps necessary) to know the solution of the GIEP (or some weak form of it) for some of the subtrees of the tree.

It is often more difficult (than giving necessary restrictions) to construct matrices $A \in \mathcal{S}(T)$ with a given, especially extremal, multiplicity list, even when that list does occur. There are three basic approaches besides ad hoc methods and computer assisted solution of equations. They are

- (a) Manipulation of polynomials, viewing the nonzero entries as variables and targeting a desired characteristic polynomial (see [Lea89] for an initial reference; this method, based on some nice formulas for the characteristic polynomial in the case of a tree (see, e.g., [Par60], [MOD89]), can be quite tedious for larger, more complicated trees).
- (b) Careful use of the implicit function theorem (initiated in [JSW]).
- (c) Division of the tree into understood parts and using the interlacing inequalities to give lower bounds that are forced to be attained by known constraints (this is along the lines of the brief discussion in Example 1 Section 34.2 involving $\Delta(T)$, but for larger trees can lead to complicated simultaneity conditions).

As an example of method (c) and its subtleties (see also Example 2 Section 34.7), consider again the tree T in Figure 34.4. Since $P(T) = 2$, the maximum multiplicity is 2, and because $d(T) = 3$, there must be at least four distinct eigenvalues, two of which have multiplicity 1. This leaves the

question of whether the list $(2, 2, 1, 1)$ (which would have to be the ordered list $(1, 2, 2, 1)$) can occur. It can, but this is the nontrivial example with smallest number of vertices. Suppose that the two multiple eigenvalues are λ and μ . We want $A \in \mathcal{S}(T)$ with $\alpha_A(\lambda) = 2$ and $\alpha_A(\mu) = 2$. Each must have a Parter vertex, which must be either vertex 3 or 4. One must be for λ (and not μ) and the other for μ (and not λ), as two consecutive eigenvalues cannot share a Parter vertex (Fact 8 section 34.1). So assume that 3 is Parter for λ and 4 for μ . Then, we must have $A[\{1\}] = \lambda = A[\{2\}]$ and $\lambda \in \sigma(A[\{4, 5, 6\}])$; and $A[\{5\}] = \mu = A[\{6\}]$ and $\mu \in \sigma(A[\{1, 2, 3\}])$. A calculation (or other methods) shows this can be achieved simultaneously.

34.6 Generalized Stars

Definitions:

A tree T in which there is at most one vertex of degree greater than two is a **generalized star**.

In a generalized star, a vertex v is a **central vertex** if its neighbors are pendant vertices of their branches, and each branch is a path. (Note that, under this definition, a path is a (degenerate) generalized star, in which any vertex is a central vertex. When referring to a path as a generalized star, one vertex has been fixed as the central vertex.)

For a central vertex v of a generalized star T , each branch of T at v is called an **arm** of T ; the **lengths** of an arm are the number of vertices in the arm.

Supposing that v is a central vertex of a generalized star T , with $\delta_T(v) = k$. Denote by $T_1, \dots, T_k$ its arms and by $l_1, \dots, l_k$ the lengths of $T_1, \dots, T_k$, respectively.

A **star** on n vertices is a tree in which there is a vertex of degree $n - 1$.

Let $u = (u_1, \dots, u_b)$, $u_1 \geq \dots \geq u_b$, and $v = (v_1, \dots, v_c)$, $v_1 \geq \dots \geq v_c$, be two nonincreasing partitions of integers M and N , respectively. If $M < N$, denote by u_e the partition of N obtained from u appending 1s to the partition u . Note that if $M = N$, then $u_e = u$.

Facts:

1. A star is trivially a generalized star.
2. If u and v are two nonincreasing partitions of integers M and N , respectively, $M \leq N$, such that $u_1 + \dots + u_s \leq v_1 + \dots + v_s$ for all s (interpreting u_s or v_s as 0 when s exceeds b or c , respectively), then trivially v majorizes u_e , denoted $u_e \leq v$. See Preliminaries.
3. [JLS03b] Let T be a generalized star on n vertices with central vertex v of degree k , $l_1, \dots, l_k$ be the lengths of the arms $T_1, \dots, T_k$, and $f(x), g_1(x), \dots, g_k(x)$ be monic polynomials with all their roots real in which $\deg f = n$, $\deg g_1 = l_1, \dots, \deg g_k = l_k$. There exists $A \in \mathcal{S}(T)$ such that A has characteristic polynomial $f(x)$ and $A[T_i]$ has characteristic polynomial $g_i(x)$ if and only if
 - Each $g_i(x)$ has only simple roots.
 - If λ is a root of $g_1(x) \cdots g_k(x)$ of multiplicity $m \geq 1$, then λ is a root of $f(x)$ of multiplicity $m - 1$.
 - The roots of $f(x)$ that are not roots of $g_1(x) \cdots g_k(x)$ are simple and strictly interlace the set of roots of $g_1(x) \cdots g_k(x)$ (multiple roots counting only once).
4. [JL02b] Let T be a generalized star on n vertices with central vertex of degree s and arm lengths $l_1 \geq \dots \geq l_s$. Then $(p_1, \dots, p_r) \in \mathcal{L}_u(T)$ if and only if
 - $\sum_{i=1}^r p_i = n$.
 - $r \geq l_1 + l_2 + 1$.
 - $p_h = p_{h+1} = \dots = p_r = 1$, in which $h = \lceil \frac{r+1}{2} \rceil$.
 - $(p_1, p_2, \dots, p_{r-l_1-1}) \leq (l_1^* - 1, \dots, l_{l_1}^* - 1)$.

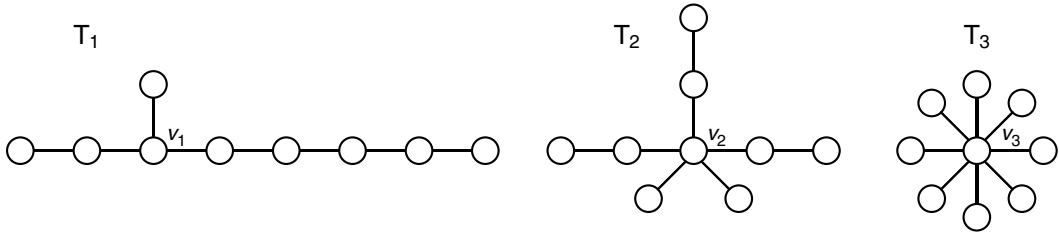


FIGURE 34.12 T_1 , T_2 , and T_3 are generalized stars on 9 vertices with central vertices v_1 , v_2 , and v_3 , respectively.

5. [JLS03b] Let T be a generalized star on n vertices with central vertex of degree s and arm lengths $l_1 \geq \dots \geq l_s$. Let $\lambda_1 < \dots < \lambda_r$ be any sequence of real numbers. Then there exists a matrix $A \in \mathcal{S}(T)$ with distinct eigenvalues $\lambda_1 < \dots < \lambda_r$ and list of ordered multiplicities $q = (q_1, \dots, q_r)$ if and only if q satisfies the following conditions:

- $\sum_{i=1}^r q_i = n$.
- If $q_i > 1$, then $1 < i < r$ and $q_{i-1} = 1 = q_{i+1}$.
- $(q_{i_1} + 1, \dots, q_{i_h} + 1)_e \preceq (l_1, \dots, l_s)^*$, in which $q_{i_1} \geq \dots \geq q_{i_h}$ are the entries of the r -tuple $(q_1, \dots, q_r)$ greater than 1.

(That is, when T is a generalized star, there is equivalence of the ordered multiplicity lists and the IEP.)

Examples:

1. Let T_1 , T_2 , and T_3 be the generalized stars in Figure 34.12. We have

$$\begin{aligned} \mathcal{L}_u(T_1) &= \{(1, 1, 1, 1, 1, 1, 1, 1), (2, 1, 1, 1, 1, 1, 1, 1)\}, \\ \mathcal{L}_u(T_2) &= \{(1, 1, 1, 1, 1, 1, 1, 1), (2, 1, 1, 1, 1, 1, 1, 1), (2, 2, 1, 1, 1, 1, 1, 1), \\ &\quad (3, 1, 1, 1, 1, 1, 1, 1), (3, 2, 1, 1, 1, 1, 1, 1), (3, 3, 1, 1, 1, 1, 1, 1), (4, 1, 1, 1, 1, 1, 1, 1), \\ &\quad (4, 2, 1, 1, 1, 1, 1, 1)\}, \end{aligned}$$

and

$$\begin{aligned} \mathcal{L}_u(T_3) &= \{(1, 1, 1, 1, 1, 1, 1, 1), (2, 1, 1, 1, 1, 1, 1, 1), (2, 2, 1, 1, 1, 1, 1, 1), \\ &\quad (3, 1, 1, 1, 1, 1, 1, 1), (3, 2, 1, 1, 1, 1, 1, 1), (3, 3, 1, 1, 1, 1, 1, 1), (4, 1, 1, 1, 1, 1, 1, 1), \\ &\quad (4, 2, 1, 1, 1, 1, 1, 1), (5, 1, 1, 1, 1, 1, 1, 1), (6, 1, 1, 1, 1, 1, 1, 1), (7, 1, 1, 1, 1, 1, 1, 1)\}. \end{aligned}$$

34.7 Double Generalized Stars

Definitions:

A **double generalized star** is a tree resulting from joining the central vertices of two generalized stars T_1 and T_2 by an edge. Such a tree will be denoted by $D(T_1, T_2)$.

A **double star** is a double generalized star $D(T_1, T_2)$ in which T_1 and T_2 are stars.

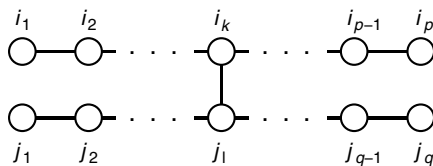


FIGURE 34.13 A double path.

A **double path** is a double generalized star $D(T_1, T_2)$ in which T_1 and T_2 are paths. When we refer to a double path T on $n = p + q$ vertices we suppose T is represented as in Figure 34.13, in which the only constraint on the connecting edge $\{i_k, j_l\}$ is that not both $k \in \{1, p\}$ and $l \in \{1, q\}$. The upper (i) path has $k - 1$ vertices to the left of the connecting vertex and another $p - k$ vertices to the right; set $s_1 = \min\{k - 1, p - k\}$, $s_2 = \min\{l - 1, q - l\}$, and $s = \min\{q, p, s_1 + s_2\}$.

Let G be a tree. Let v be a vertex of G of degree k and $G_1, \dots, G_k$ be the components of $G \setminus \{v\}$ having order $l_1, \dots, l_k$, respectively. To the tree G is **associated the generalized star**, $S_v(G)$, with central vertex v of degree k , and with arms $T_1, \dots, T_k$ of lengths $l_1, \dots, l_k$, respectively.

Let u_1 and u_2 be adjacent vertices of a tree G . Denote by G_{u_1} the connected component of $G \setminus \{u_2\}$ that contains u_1 and by G_{u_2} the connected component of $G \setminus \{u_1\}$ that contains u_2 . Put $S_1 = S_{u_1}(G_{u_1})$ and $S_2 = S_{u_2}(G_{u_2})$. Now, to the tree G is **associated the double generalized star** $D(S_1, S_2)$, which is denoted by $D_{u_1, u_2}(G)$.

Given a vertex v of a tree T and an eigenvalue λ of a matrix $A \in \mathcal{S}(T)$, λ is an **upward eigenvalue** of A at v if $\alpha_{A(v)}(\lambda) = \alpha_A(\lambda) + 1$, and $\alpha_A(\lambda)$ is an **upward multiplicity** of A at v .

If $q = q(A) = (q_1, \dots, q_r)$ is the list of ordered multiplicities of A , define the **list of upward multiplicities** of A at v , denoted by $\hat{q}$, as the list with the same entries as q but in which any upward multiplicity q_i of A at v is marked as $\hat{q}_i$ in $\hat{q}$.

Given a generalized star T with central vertex v , we denote by $\hat{\mathcal{L}}_o(T)$ the **set of all lists of upward multiplicities** at v occurring among matrices in $\mathcal{S}(T)$.

Facts:

1. A double path is a tree whose path cover number is 2.
2. [JLS03b] Let T be a generalized star on n vertices with central vertex v of degree k and arm lengths $l_1 \geq \dots \geq l_k$. Let $\lambda_1 < \dots < \lambda_r$ be any sequence of real numbers. Then there exists a matrix A in $\mathcal{S}(T)$ with distinct eigenvalues $\lambda_1 < \dots < \lambda_r$ and a list of upward multiplicities $\hat{q} = (\hat{q}_1, \dots, \hat{q}_r)$ if and only if $\hat{q}$ satisfies the following conditions:
 - $\sum_{i=1}^r q_i = n$.
 - If q_i is an upward multiplicity in $\hat{q}$, then $1 < i < r$ and neither q_{i-1} nor q_{i+1} is an upward multiplicity in $\hat{q}$.
 - $(q_{i_1} + 1, \dots, q_{i_h} + 1)_e \leq (l_1, \dots, l_k)^*$, in which $q_{i_1} \geq \dots \geq q_{i_h}$ are the upward multiplicities of $\hat{q}$.
3. [JLS03b] (Superposition Principle) Let $D(T_1, T_2)$ be a double generalized star, $\hat{b} = (\widehat{b_1, \dots, b_{s_1}}) \in \hat{\mathcal{L}}_o(T_1)$, and $\hat{c} = (\widehat{c_1, \dots, c_{s_2}}) \in \hat{\mathcal{L}}_o(T_2)$. Construct any $b^+ = (b_1^+, \dots, b_{s_1+t_1}^+)$ and $c^+ = (c_1^+, \dots, c_{s_2+t_2}^+)$ subject to the following conditions:
 - $t_1, t_2 \in \mathbb{N}_0$ and $s_1 + t_1 = s_2 + t_2$.
 - b^+ (respectively, c^+) is obtained from $\hat{b}$ (respectively, $\hat{c}$) by inserting t_1 (respectively, t_2) 0s.
 - b_i^+ and c_i^+ cannot both be 0.
 - If $b_i^+ > 0$ and $c_i^+ > 0$, then at least one of the b_i^+ or c_i^+ must be an upward multiplicity of $\hat{b}$ or $\hat{c}$.

Then $b^+ + c^+ \in \mathcal{L}_o(D(T_1, T_2))$. Moreover, $a \in \mathcal{L}_o(D(T_1, T_2))$ if and only if there are $\hat{b} \in \hat{\mathcal{L}}_o(T_1)$, $\hat{c} \in \hat{\mathcal{L}}_o(T_2)$ such that $a = b^+ + c^+$.

4. Let T be a tree, v be a vertex of T , and v_1, v_2 be adjacent vertices of T . Then

- $\mathcal{L}_u(S_v(T)) \subseteq \mathcal{L}_u(T)$, $\mathcal{L}_o(S_v(T)) \subseteq \mathcal{L}_o(T)$.
- $\mathcal{L}_u(D_{v_1, v_2}(T)) \subseteq \mathcal{L}_u(T)$, $\mathcal{L}_o(D_{v_1, v_2}(T)) \subseteq \mathcal{L}_o(T)$.

5. [JL02b] Let T be a double path on $n = p + q$ vertices and suppose that $A \in \mathcal{S}(T)$. Then

- The maximum multiplicity of an eigenvalue of A is 2.
- The diameter of G is $\max\{p, q, p + q - (s_1 + s_2)\} - 1$, so that A has at least $\max\{p, q, p + q - (s_1 + s_2)\}$ distinct eigenvalues.
- A has at most s multiplicity 2 eigenvalues.
- The possible list of unordered multiplicities for T , $\mathcal{L}_u(T)$, consists of all partitions of $p + q$ into parts each one not greater than two and with at most s equal to 2.
- Any list in $\mathcal{L}_u(T)$ has at least $n - 2s$ 1s.

Examples:

1. Let T_1 and T_2 be the stars in Figure 34.14 with central vertices v_1 and v_2 , respectively, and G be the double star $D(T_1, T_2)$. By Fact 2, we have that

$$\hat{\mathcal{L}}_o(T_1) = \{(1, \hat{2}, 1), (1, \hat{1}, 1, 1), (1, 1, \hat{1}, 1), (1, 1, 1, 1)\}$$

and

$$\hat{\mathcal{L}}_o(T_2) = \{(1, \hat{1}, 1), (1, 1, 1)\}.$$

Applying the Superposition Principle (Fact 3) to the lists of upward multiplicities of T_1 and T_2 , it follows that

$$\begin{aligned} \mathcal{L}_o(G) = & \{(1, 3, 2, 1), (1, 2, 3, 1), (1, 3, 1, 1, 1), (1, 1, 3, 1, 1), (1, 1, 1, 3, 1), \\ & (1, 2, 2, 1, 1), (1, 2, 1, 2, 1), (1, 1, 2, 2, 1), (1, 2, 1, 1, 1, 1), \\ & (1, 1, 2, 1, 1, 1), (1, 1, 1, 2, 1, 1), (1, 1, 1, 1, 2, 1), (1, 1, 1, 1, 1, 1, 1)\}. \end{aligned}$$

For example, $(1, 3, 2, 1) \in \mathcal{L}_o(G)$ because $\hat{b} = (1, \hat{2}, 1) \in \hat{\mathcal{L}}_o(T_1)$, $\hat{c} = (1, \hat{1}, 1) \in \hat{\mathcal{L}}_o(T_2)$, and

$$(1, 3, 2, 1) = b^+ + c^+ = (1, \hat{2}, 1, 0) + (0, 1, \hat{1}, 1).$$

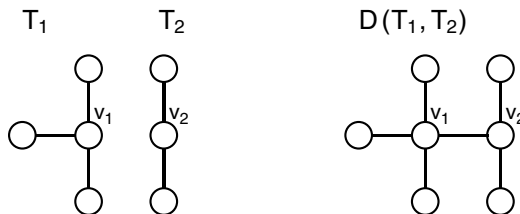


FIGURE 34.14 Stars and a double star.

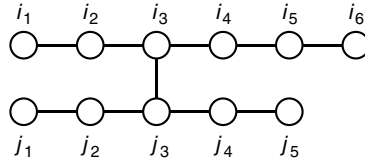


FIGURE 34.15 A double path on 11 vertices.

2. Consider the double path T on 11 vertices in Figure 34.15. Let T_1 be the path with vertices $i_1, \dots, i_6$ and T_2 be the path with vertices $j_1, \dots, j_5$ (subgraphs of T induced by the mentioned vertices). Because T_1 and T_2 are generalized stars with central vertices i_3 and j_3 , respectively, from Fact 2 we conclude that $\hat{b} = (1, \hat{1}, 1, \hat{1}, 1, 1) \in \hat{\mathcal{L}}_o(T_1)$ and $\hat{c} = (1, \hat{1}, 1, \hat{1}, 1) \in \hat{\mathcal{L}}_o(T_2)$. Since, for example,

$$(1, 2, 2, 2, 2, 1, 1) = b^+ + c^+ = (0, 1, \hat{1}, 1, \hat{1}, 1, 1) + (1, \hat{1}, 1, \hat{1}, 1, 0, 0),$$

by the Superposition Principle, we conclude that $(1, 2, 2, 2, 2, 1, 1) \in \mathcal{L}_o(T)$ and, therefore, $(2, 2, 2, 2, 1, 1, 1) \in \mathcal{L}_u(T)$. We may construct a matrix $A \in \mathcal{S}(T)$ with list of multiplicities $(2, 2, 2, 2, 1, 1, 1)$ in the following way.

Pick real numbers $\lambda_1 > \lambda_2 > \mu_2 > \lambda_3 > \mu_3 > \lambda_4$. Construct A_1 with graph T_1 such that A_1 has eigenvalues $\lambda_1, \mu_1, \lambda_2, \mu_2, \lambda_3, \lambda_4$ and such that the eigenvalues of $A_1[\{i_1, i_2\}]$ and $A_1[\{i_4, i_5, i_6\}]$ are μ_1, μ_2 and μ_1, μ_2, μ_3 , respectively; construct A_2 with graph T_2 such that A_2 has eigenvalues $\mu_1, \lambda_2, \mu_2, \lambda_3, \mu_3$ and such that the eigenvalues of both $A_2[\{j_1, j_2\}]$ and $A_2[\{j_4, j_5\}]$ are λ_2, λ_3 . According to Fact 3 section 34.6, these constructions are possible. Now construct A with graph T and such that $A[T_1] = A_1$ and $A[T_2] = A_2$. Then i_3 is a strong Parter vertex for μ_1, μ_2 , while j_3 is a strong Parter for λ_2, λ_3 and so $(2, 2, 2, 2, 1, 1, 1)$ is the list of unordered multiplicities of A .

3. Regarding Fact 4, the results for generalized stars or double generalized stars may be extended to a general tree T by associating with T either a generalized star or a double generalized star according to the given definitions. (See also [JLS03b, Theorem 10] for a corresponding result for the GIEP.)

Note that, under our definition, there are many different possibilities to associate either a generalized star or a double generalized star to a given tree T , and so Fact 3 provides many possible lists for $\mathcal{L}(T)$. The natural question is to ask whether all the elements of $\mathcal{L}(T)$ can be obtained in this manner. The answer is no. It suffices to note that the path cover number of T will be, in general, strictly greater than that of either of $S_v(T)$ or $D_{v_1, v_2}(T)$ for any possible choice of v, v_1 , and v_2 . So any lists for which the maximum multiplicity occurs cannot generally be obtained from the inclusions in Fact 4. For example, the tree T in Figure 34.16 has path cover number 3, which is strictly greater than the maximum path cover number 2, of any generalized star or double generalized star associated with T .

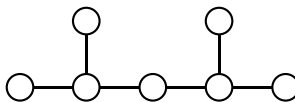


FIGURE 34.16 A tree with path cover number 3.

34.8 Vines

Definitions:

A **binary tree** is a tree in which no vertex has degree greater than 3.

A **vine** is a binary tree in which every degree 3 vertex is adjacent to at least one vertex of degree 1 and no two vertices of degree 3 are adjacent.

Facts:

1. [JSW] Let T be a vine on n vertices. The set $\mathcal{L}_u(T)$ consists of all sequences that are majorized by the sequence $s = (P(T), 1, \dots, 1)$ (s being a partition of n).
(The description of $\mathcal{L}_u(T)$ was given by using the implicit function theorem technique referred to in section 34.5.)

References

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35

Matrix Completion Problems

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35.1	Introduction	35-2
35.2	Positive Definite and Positive Semidefinite Matrices	35-8
35.3	Euclidean Distance Matrices	35-9
35.4	Completely Positive and Doubly Nonnegative Matrices	35-10
35.5	Copositive and Strictly Copositive Matrices	35-11
35.6	M - and M_0 -Matrices	35-12
35.7	Inverse M -Matrices	35-14
35.8	P -, $P_{0,1}$ -, and P_0 -Matrices	35-15
35.9	Positive P -, Nonnegative P -, Nonnegative $P_{0,1}$ -, and Nonnegative P_0 -Matrices	35-17
35.10	Entry Sign Symmetric P -, Entry Sign Symmetric P_0 -, Entry Sign Symmetric $P_{0,1}$ -, Entry Weakly Sign Symmetric P -, and Entry Weakly Sign Symmetric P_0 -Matrices	35-19
	References	35-20

A partial matrix is a rectangular array of numbers in which some entries are specified while others are free to be chosen. A completion of a partial matrix is a specific choice of values for the unspecified entries. A matrix completion problem asks whether a partial matrix (or family of partial matrices with a given pattern of specified entries) has a completion of a specific type, such as a positive definite matrix. In some cases, a “best” completion is sought.

Matrix completion problems arise in applications whenever a full set of data is not available, but it is known that the full matrix of data must have certain properties. Such applications include molecular biology and chemistry (see Chapter 60), seismic reconstruction problems, mathematical programming, and data transmission, coding, and image enhancement problems in electrical and computer engineering.

A matrix completion problem for a family of partial matrices with a given pattern of specified entries is usually studied by means of graphs or digraphs. If a pattern of specified entries does not always allow completion to the desired type of matrix, conditions on the entries that will allow such completion are sought. A question of finding the “best completion” often involves optimization techniques. Matrix completion results are usually constructive, with the result established by giving a specific construction for a completion.

In this chapter, we focus on completion problems involving classes of matrices that generalize the positive definite matrices, and emphasize graph theoretic techniques that allow completion of families of matrices. This chapter is organized by class of matrices, with the symmetric classes first. The authors also maintain a Web page containing updated information [HW].

Organizing the information by classes of matrices provides easy access to results about a particular class, but obscures techniques that apply to the matrix completion problems of many classes and relationships between completion problems for different classes. For information on these subjects, see for example [FJT00], [Hog01], and [Hog03a].

35.1 Introduction

All matrices and partial matrices discussed here are square. Graphs allow loops but do not allow multiple edges. The definitions and terminology about graphs and digraphs given in Chapter 28 and Chapter 29 is used here; however, the association between matrices and digraphs is different from the association in those chapters, where an arc is associated with a nonzero entry. Here an arc is associated with a specified entry.

Definitions:

A **partial matrix** is a square array in which some entries are specified and others are not. An unspecified entry is denoted by ? or by x_{ij} . An ordinary matrix is considered a partial matrix, as is a matrix with no specified entries.

A **completion** of a partial matrix is a choice of values for the unspecified entries.

A partial matrix B is **combinatorially symmetric** (also called **positionally symmetric**) if b_{ij} specified implies b_{ji} specified.

Let B be an $n \times n$ partial matrix. The **digraph** of B , $\mathcal{D}(B) = (V, E)$, has vertices $V = \{1, \dots, n\}$, and for each i and j in V , the arc $(i, j) \in E$ exactly when b_{ij} is specified.

Let B be an $n \times n$ combinatorially symmetric partial matrix. The **graph** of B , $\mathcal{G}(B) = (V, E)$, has vertices $V = \{1, \dots, n\}$, and for each i and j in V , the edge $\{i, j\} \in E$ exactly when b_{ij} is specified.

A connected graph or digraph is **nonseparable** if it does not have a cut-vertex.

A **block** of a graph or digraph is a maximal nonseparable sub(di)graph. This use of “block” is for graphs and digraphs and differs from “block” in a block matrix.

A graph (respectively, digraph) is a **clique** if every vertex has a loop and for any two distinct vertices u, v , the edge $\{u, v\}$ is present (respectively, both arcs $(u, v), (v, u)$ are present).

A graph or digraph is **block-clique** (also called **1-chordal**) if every block is a clique.

A digraph $G = (V, E)$ is **symmetric** if $(i, j) \in E$ implies $(j, i) \in E$ for all $i, j \in V$.

A digraph $G = (V, E)$ is **asymmetric** if $(i, j) \in E$ implies $(j, i) \notin E$ for all distinct $i, j \in V$.

A **simple cycle** in a digraph is an induced subdigraph that is a cycle.

A digraph (respectively, graph) G has the **X -completion property** (where X is a type of matrix) if every partial X -matrix B such that $\mathcal{D}(B) = G$ (respectively, $\mathcal{G}(B) = G$) can be completed to an X -matrix. In the literature, the phrase “has X -completion” is sometimes used for “has the X -completion property.”

A class X is **closed under permutation similarity** if whenever A is an X -matrix and P is a permutation matrix, then $P^T A P$ is an X -matrix.

A class X is **hereditary** (or **closed under taking principal submatrices**) if whenever A is an X -matrix and $\alpha \subseteq \{1, \dots, n\}$, then $A[\alpha]$ is an X -matrix.

A class X is **closed under matrix direct sums** if whenever $A_1, A_2, \dots, A_k$ are X -matrices, then $A_1 \oplus A_2 \oplus \dots \oplus A_k$ is an X -matrix.

A class X has the **triangular property** if whenever A is a block triangular matrix and every diagonal block is an X -matrix, then A is an X matrix.

A partial matrix B is in **pattern block triangular form** if the adjacency matrix of $\mathcal{D}(B)$ is in block triangular form.

Note: Many matrix terms, such as size, entry, submatrix, etc., are applied in the obvious way to partial matrices.

Facts:

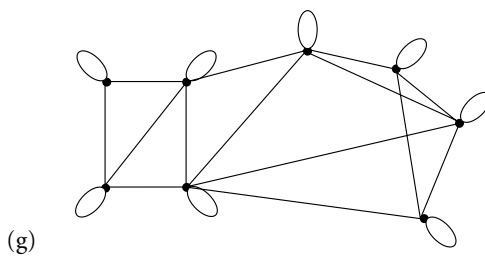
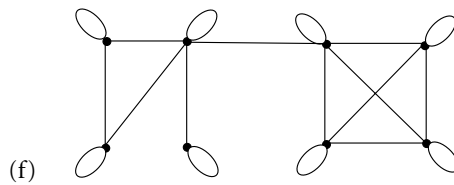
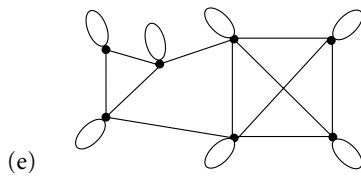
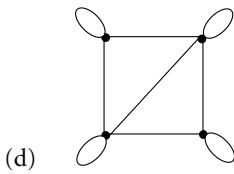
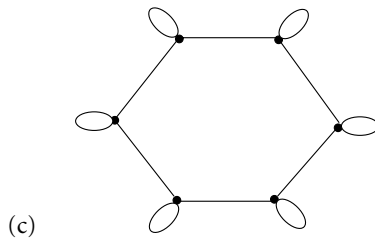
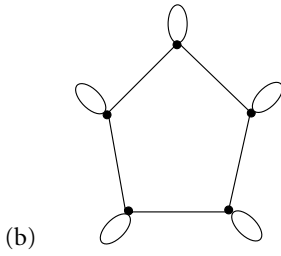
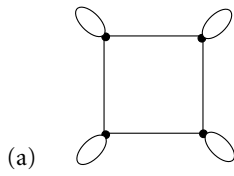
Let X be one of the following classes of matrices: positive (semi)definite matrices, Euclidean distance matrices, (symmetric) M -matrices, (symmetric) M_0 -matrices, (symmetric) inverse M -matrices, completely positive matrices, doubly nonnegative matrices, (strictly) copositive matrices, P -matrices, P_0 -matrices, $P_{0,1}$ -matrices, nonnegative P -matrices, nonnegative P_0 -matrices, positive P -matrices, entry (weakly) sign symmetric P -matrices, entry (weakly) sign symmetric P_0 -matrices, entry (weakly) sign symmetric $P_{0,1}$ -matrices. (The definitions of these classes can be found in the relevant sections.)

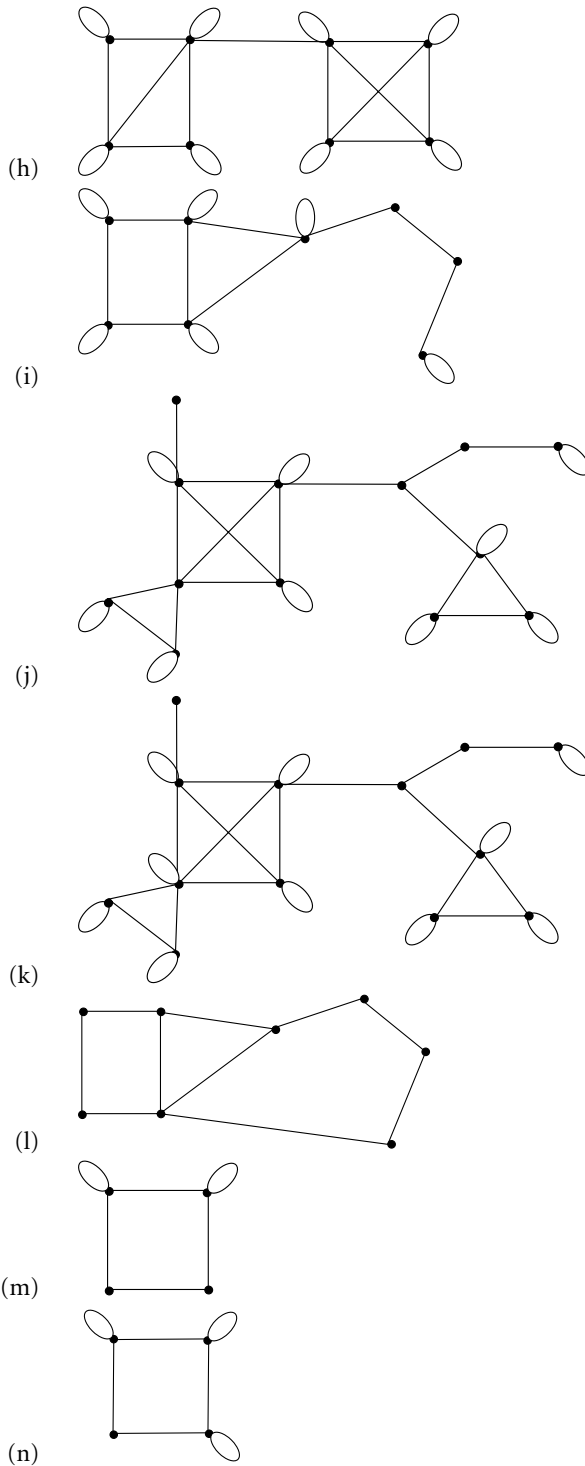
Proofs of the facts below can be found in [Hog01] for most of the classes discussed, and the proofs given there apply to all the classes X listed above. Most of these facts are also in the original papers discussing the completion problem for a specific class; those references are listed in the section devoted to the class. In the literature, when it is assumed that a partial matrix B has every diagonal entry specified (equivalently, every vertex of the graph $\mathcal{G}(B)$ or digraph $\mathcal{D}(B)$ has a loop), it is customary to suppress all the loops and treat $\mathcal{G}(B)$ or $\mathcal{D}(B)$ as a simple graph or digraph. That is not done in this chapter because of the danger of confusion. Also, in some references, such as [Hog01], a mark is used to indicate a specified vertex instead of a loop. This has no effect on the results (but requires translation of the notation). If X is a symmetric class of matrices, then there is no loss of generality in assuming every partial matrix is combinatorially symmetric and this assumption is standard practice.

1. If B is a combinatorially symmetric partial matrix, then $\mathcal{D}(B)$ is a symmetric digraph and $\mathcal{G}(B)$ is the graph associated with $\mathcal{D}(B)$. Combinatorially symmetric partial matrices are usually studied by means of graphs rather than digraphs, and it is understood that the graph represents the associated symmetric digraph.
2. Each of the classes X listed at the beginning of the facts is closed under permutation similarity. This fact is not true for the classes of totally nonnegative and totally positive matrices. (See [FJS00] for information about matrix completion problems for these matrices.)
3. Applying a permutation similarity to a partial matrix B corresponds to renumbering the vertices of the digraph $\mathcal{D}(B)$ (or graph $\mathcal{G}(B)$ if B is combinatorially symmetric).
4. Renumbering the vertices of a graph or digraph does not affect whether it has the X -completion property. It is customary to use unlabeled (di)graph diagrams. This fact is not true for the classes of totally nonnegative and totally positive matrices.
5. Each of the classes X listed at the beginning of the facts is hereditary.
6. Let B be a partial matrix and $\alpha \subseteq \{1, \dots, n\}$. The digraph of the principal submatrix $B[\alpha]$ is isomorphic to the subdigraph of $\mathcal{D}(B)$ induced by α (and is customarily identified with it). The same is true for the graph if B is combinatorially symmetric.
7. If a graph or digraph G has the X -completion property, then every induced subgraph or induced subdigraph of G has the X -completion property.
8. Each of the classes X listed at the beginning of the facts is closed under matrix direct sums.
9. Let B be a partial matrix such that all specified entries are contained in diagonal blocks $B_1, B_2, \dots, B_k$. The connected components of $\mathcal{D}(B)$ are isomorphic to the $\mathcal{D}(B_i)$, $i = 1, \dots, k$. The same is true for $\mathcal{G}(B)$ if B is combinatorially symmetric.
10. A graph or digraph G has the X -completion property if and only if every connected component of G has the X -completion property.
11. If X has the triangular property, B is a partial matrix in pattern block triangular form, and each pattern diagonal block can be completed to an X -matrix, then B can be completed to an X -matrix.
12. If X has the triangular property and is closed under permutation similarity, then a graph or digraph G has the X -completion property if and only if every strongly connected component of G has the X -completion property.
13. A block-clique graph is chordal.
14. A block-clique digraph is symmetric.

Examples:

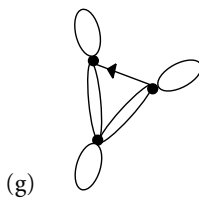
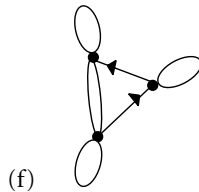
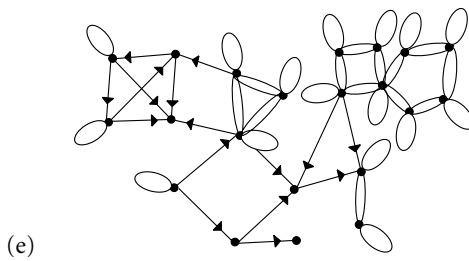
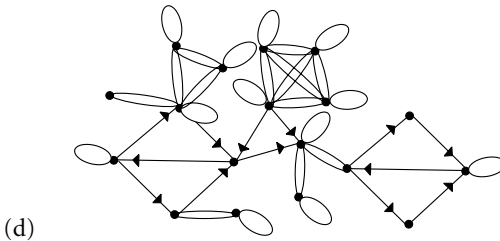
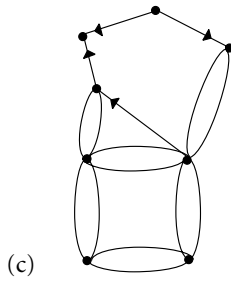
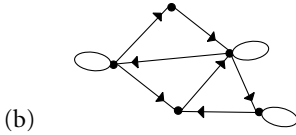
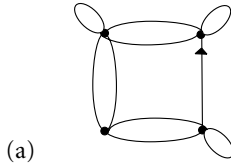
1. Graphs (a) through (n) will be used in the examples in the following sections.

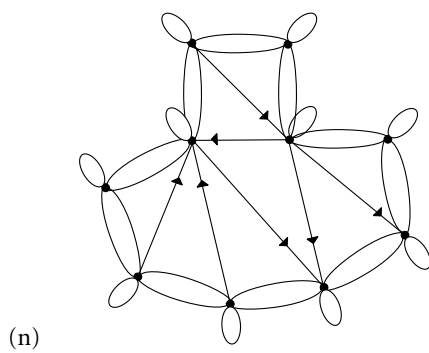
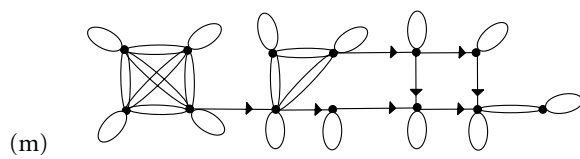
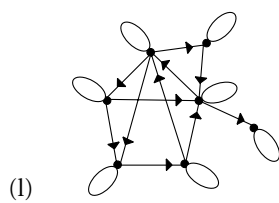
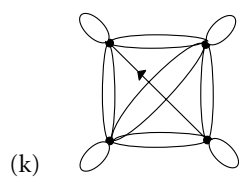
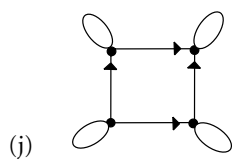
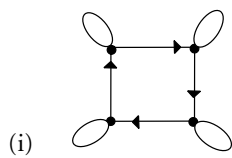
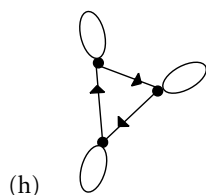


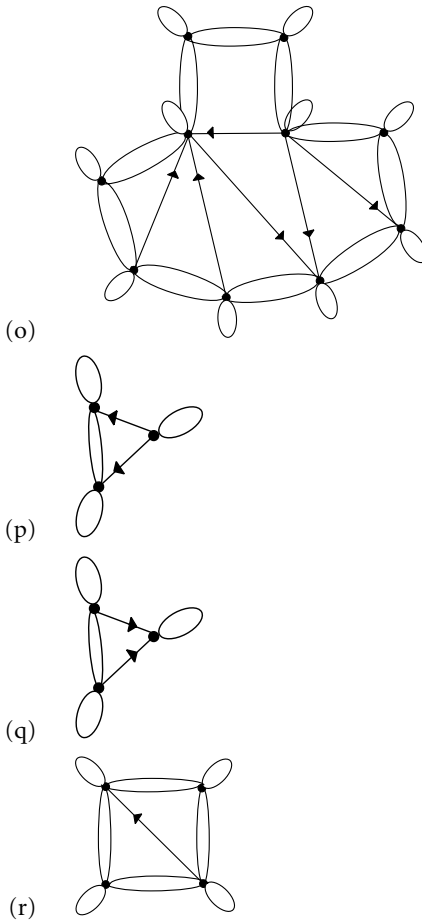


2. The matrix $\begin{bmatrix} 1 & 3 & ? & 0 \\ -1 & 1 & -7 & ? \\ ? & -7 & 1 & 2 \\ 8 & ? & 0 & 1 \end{bmatrix}$ is a partial matrix specifying the graph 1a with vertices numbered 1, 2, 3, 4 clockwise from upper left (or any other numbering around the cycle in order).

3. The graph 1f is block-clique, and this is the only block-clique graph in Example 1.
4. The following digraphs ((a) through (r)) will be used in the examples in the following sections. Note that when both arcs (i, j) and (j, i) are present, the arrows are omitted.







5. None of the digraphs in Example 4 are symmetric. (We diagram a symmetric digraph by its associated graph.) Digraphs 4b, 4h, 4i, 4j, and 4l are asymmetric.

35.2 Positive Definite and Positive Semidefinite Matrices

In this section, all matrices are real or complex.

Definitions:

The matrix A is **positive definite** (respectively, **positive semidefinite**) if A is Hermitian and for all $\mathbf{x} \neq 0$, $\mathbf{x}^* A \mathbf{x} > 0$ (respectively, $\mathbf{x}^* A \mathbf{x} \geq 0$).

The partial matrix B is a **partial positive definite matrix** (respectively, **partial positive semidefinite matrix**) if every fully specified principal submatrix of B is a positive definite matrix (respectively, positive semidefinite matrix), and whenever b_{ij} is specified then so is b_{ji} and $b_{ji} = \overline{b_{ij}}$.

Facts:

1. A Hermitian matrix A is positive definite (respectively, positive semidefinite) if and only if it is positive stable (respectively, positive semistable) if and only if all principal minors are positive (respectively, nonnegative). There are many additional characterizations. (See Chapter 8.4 for more information.)
2. [GJS84] A graph that has a loop at every vertex has the positive definite (positive semidefinite) completion property if and only if it is chordal. (For information on how to construct such a completion, see [GJS84] and [DG81].)

3. [GJS84] A graph has the positive definite completion property if and only if the subgraph induced by the vertices with loops has the positive definite completion property.
4. [Hog01] A graph G has the positive semidefinite completion property if and only if for each connected component H of G , either H has a loop at every vertex and is chordal, or H has no loops.
5. [GJS84] If B is a partial positive definite matrix with all diagonal entries specified such that $\mathcal{G}(B)$ is chordal, then there is a unique positive definite completion A of B that maximizes the determinant, and this completion has the property that whenever the i, j -entry of B is unspecified, the i, j -entry of A^{-1} is zero.
6. [Fie66] Let C be a partial positive semidefinite matrix such that every diagonal entry is specified and $\mathcal{G}(B)$ with loops suppressed is a cycle. If any diagonal entry is 0, then B can be completed to a positive semidefinite matrix. If every diagonal entry is nonzero, there is a positive diagonal matrix D such that every diagonal entry of $C = DBD$ is equal to 1. Let the specified off-diagonal entries of C be denoted $c_1, \dots, c_n$. Then C (and, hence, B) can be completed to a positive semidefinite matrix if and only if the following *cycle conditions* are satisfied:

$$\begin{aligned} 2 \max_{1 \leq k \leq n} \arccos|c_k| &\leq \sum_{k=1}^n \arccos|c_k| && \text{for } c_1 \dots c_n > 0, \\ \sum_{k=1}^n \arccos|c_k| &\geq \pi && \text{for } c_1 \dots c_n \leq 0. \end{aligned}$$

(See [BJL96] for additional information.)

Examples:

The graphs for the examples can be found in Example 1 of Section 35.1.

1. The graphs 1d, 1f, and 1h have both the positive definite and positive semidefinite completion properties by Fact 2.
2. The graphs 1a, 1b, 1c, 1e, and 1g have neither the positive definite nor the positive semidefinite completion property by Fact 2.
3. The graphs 1j, 1k, 1l, 1m, and 1n have the positive definite completion property by Facts 3 and 2.
4. The graph 1i does not have the positive definite completion property by Facts 3 and 2.
5. The graph 1l has the positive semidefinite completion property by Fact 4.
6. The graphs 1i, 1j, 1k, 1m, and 1n do not have the positive semidefinite completion property by Fact 4.

7. The partial matrix $B = \begin{bmatrix} 1 & .3 & ? & ? & -.1 \\ .3 & 1 & 1 & ? & ? \\ ? & 1 & 1 & .2 & ? \\ ? & ? & .2 & 1 & 1 \\ -.1 & ? & ? & 1 & 1 \end{bmatrix}$ can be completed to a positive semidefinite

matrix by Fact 6 because

$$\sum_{k=1}^n \arccos|c_k| = 4.10617 \geq \pi.$$

35.3 Euclidean Distance Matrices

In this section, all matrices are real.

Definitions:

The matrix $A = [a_{ij}]$ is a **Euclidean distance matrix** if there exist vectors $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ (for some $d \geq 1$) such that $a_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$ for all $i, j = 1, \dots, n$.

The partial matrix B is a **partial Euclidean distance matrix** if every diagonal entry is specified and equal to 0, every fully specified principal submatrix of B is a Euclidean distance matrix, and whenever b_{ij} is specified then so is b_{ji} and $b_{ji} = b_{ij}$.

Facts:

1. Every Euclidean distance matrix has all diagonal elements equal to 0. There is no loss of generality by considering only a graph that has loops at every vertex, and requiring all diagonal entries of partial Euclidean distance matrices to be 0.
2. [Lau98] A graph with a loop at every vertex has the Euclidean distance completion property if and only if it is chordal.
3. [Lau98] A graph with a loop at every vertex has the Euclidean distance completion property if and only if it has the positive semidefinite completion property. There is a method for transforming the Euclidean distance completion problem into the positive semidefinite completion problem via the Schoenberg transform that provides additional information about conditions on entries that are sufficient to guarantee completion.

Examples:

The graphs for the examples can be found in Example 1 of Section 35.1.

1. The graphs 1d, 1f, and 1h have the Euclidean distance completion property by Fact 2.
2. The graphs 1a, 1b, 1c, 1e, and 1g do not have the Euclidean distance completion property by Fact 2.

35.4 Completely Positive and Doubly Nonnegative Matrices

In this section, all matrices are real.

Definitions:

The matrix A is a **completely positive matrix** if $A = CC^T$ for some nonnegative $n \times m$ matrix C .

A matrix is a **doubly nonnegative matrix** if it is positive semidefinite and every entry is nonnegative.

The partial matrix B is a **partial completely positive matrix** (respectively, **partial doubly nonnegative matrix**) if every fully specified principal submatrix of B is a completely positive matrix (respectively, doubly nonnegative matrix), and whenever b_{ij} is specified then so is b_{ji} and $b_{ji} = b_{ij}$, and all specified off-diagonal entries are nonnegative.

Facts:

1. A completely positive matrix is doubly nonnegative.
2. [DJ98] A graph that has a loop at every vertex has the completely positive completion property (respectively, doubly nonnegative completion property) if and only if it is block-clique.
3. [Hog02] A graph G has the completely positive completion property (respectively, doubly nonnegative completion property) if and only if for every connected component H of G , H is block-clique, or H has no loops.
4. A graph has the completely positive completion property if and only if it has the doubly nonnegative completion property.
5. [DJK00] A partial matrix that satisfies the conditions of Fact 6 of Section 35.2 can be completed to a CP- (respectively, DN-) matrix.

Examples:

The graphs for the examples can be found in Example 1 of Section 35.1.

1. The graph 1f has both the completely positive completion property and the doubly nonnegative completion property by Fact 2.

2. The graphs 1a, 1b, 1c, 1d, 1e, 1g, and 1h have neither the completely positive completion property nor the doubly nonnegative completion property by Fact 2.
3. The graph 1l has both the completely positive completion property and the doubly nonnegative completion property by Fact 3.
4. The graphs 1i, 1j, 1k, 1m, and 1n have neither the completely positive completion property nor the doubly nonnegative completion property by Fact 3.

35.5 Copositive and Strictly Copositive Matrices

In this section, all matrices are real.

Definitions:

The symmetric matrix A is **strictly copositive** if $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \geq 0$ and $\mathbf{x} \neq 0$; A is **copositive** if $\mathbf{x}^T A \mathbf{x} \geq 0$ for all $\mathbf{x} \geq 0$.

The partial matrix B is a **partial strictly copositive matrix** (respectively, **partial copositive matrix**) if every fully specified principal submatrix of B is a strictly copositive matrix (respectively, copositive matrix) and whenever b_{ij} is specified then so is b_{ji} and $b_{ji} = b_{ij}$.

Facts:

1. If A is (strictly) copositive, then so is $A + M$ for any symmetric nonnegative matrix M .
2. [HJR05] [Hog] Every partial strictly copositive matrix can be completed to a strictly copositive matrix using the method described in Facts 4 and 5 below.
3. [HJR05] Every partial copositive matrix that has every diagonal entry specified can be completed to a copositive matrix using the completion described in Fact 4 below. There exists a partial copositive matrix with an unspecified diagonal entry that cannot be completed to a copositive matrix (see Example 2 below).
4. [HJR05] Let B be a partial copositive matrix with every diagonal entry specified. For each pair of unspecified off-diagonal entries, set $x_{ij} = x_{ji} = \sqrt{b_{ii}b_{jj}}$. The resulting matrix is copositive, and is strictly copositive if B is a partial strictly copositive matrix.
5. [Hog] Any completion of a partial strictly copositive matrix omitting only one diagonal entry found by Algorithm 1 is a strictly copositive matrix. If B is a partial strictly copositive matrix that omits some diagonal entries, values for these entries can be chosen one at a time using Algorithm 1, using the largest value obtained by considering all principal submatrices that are completed by the choice of that diagonal entry, to obtain a partial strictly copositive matrix with specified diagonal that agrees with B on every specified entry of B .

Algorithm 1: Completing one unspecified diagonal entry

Let $B = \begin{bmatrix} x_{11} & \mathbf{b}^T \\ \mathbf{b} & B_1 \end{bmatrix}$ be a partial strictly copositive $n \times n$ matrix having all entries

except the 1,1-entry specified. Let $\|\cdot\|$ be a vector norm.

Complete B by choosing a value for x_{11} as follows:

1. $\beta = \min_{\mathbf{y} \in \mathbb{R}^{n-1}, \mathbf{y} \geq 0, \|\mathbf{y}\|=1} \mathbf{b}^T \mathbf{y}$.
2. $\gamma = \min_{\mathbf{y} \in \mathbb{R}^{n-1}, \mathbf{y} \geq 0, \|\mathbf{y}\|=1} \mathbf{y}^T B_1 \mathbf{y}$.
3. $x_{11} > \frac{\beta^2}{\gamma}$.

6. [HJR05], [Hog] Every graph has the strictly copositive completion property.
7. [HJR05], [Hog] A graph has the copositive completion property if and only if for each connected component H of G , either H has a loop at every vertex, or H has no loops.

Examples:

1. The partial matrix $B = \begin{bmatrix} x_{11} & -5 & 1 & x_{14} & x_{15} & x_{16} \\ -5 & 1 & -2 & x_{24} & x_{25} & 1 \\ 1 & -2 & 5 & 1 & -1 & -1 \\ x_{14} & x_{24} & 1 & 1 & x_{45} & 1 \\ x_{15} & x_{25} & -1 & x_{45} & x_{55} & x_{56} \\ x_{16} & 1 & -1 & 1 & x_{56} & 3 \end{bmatrix}$ is a partial strictly copositive matrix.

We use the method in Facts 4 and 5 to complete B to a strictly copositive matrix:

Select index 5. The only principal submatrix completed by a choice of b_{55} is $B[\{3, 5\}]$. Any value that makes $x_{55}b_{33} > b_{35}^2$ will work; we choose $x_{55} = 1$.

Select index 1. The only principal submatrices completed by a choice of b_{11} are principal submatrices

of $B[\{1, 2, 3\}] = \begin{bmatrix} x_{11} & \mathbf{b}^T \\ \mathbf{b} & B[\{2, 3\}] \end{bmatrix}$. Apply Algorithm 1 (using $\|\cdot\|_1$):

1. $\beta = \min_{\|y\|_1=1} \mathbf{b}^T y = -5$.
2. $\gamma = \min_{\|y\|_1=1} y^T B[\{2, 3\}]y = \frac{1}{10}$.
3. Choose $x_{11} > \frac{\beta^2}{\gamma}$; we choose $b_{11} = 256$.

Then by Fact 4, $B = \begin{bmatrix} 256 & -5 & 1 & 16 & 16 & 16\sqrt{3} \\ -5 & 1 & -2 & 1 & 1 & 1 \\ 1 & -2 & 5 & 1 & -1 & -1 \\ 16 & 1 & 1 & 1 & 1 & 1 \\ 16 & 1 & -1 & 1 & 1 & \sqrt{3} \\ 16\sqrt{3} & 1 & -1 & 1 & \sqrt{3} & 3 \end{bmatrix}$ is a strictly copositive matrix.

2. $B = \begin{bmatrix} x_{11} & -1 \\ -1 & 0 \end{bmatrix}$ is a partial copositive matrix that cannot be completed to a copositive matrix

because once x_{11} is chosen (clearly $x_{11} > 0$), then $\mathbf{v} = [\frac{1}{x_{11}}, 1]^T$ results in $\mathbf{v}^T B \mathbf{v} = \frac{-1}{x_{11}} < 0$.

3. The graphs 1a, 1b, 1c, 1d, 1e, 1f, 1g, 1h, and 1i have copositive completion by Fact 7, and these are the only graphs in Example 1 of section 35.1 that have the copositive completion property.

35.6 M - and M_0 -Matrices

In this section, all matrices are real.

Definitions:

The matrix A is an **M -matrix** (respectively, **M_0 -matrix**) if there exist a nonnegative matrix P and a real number $s > \rho(P)$ (respectively, $s \geq \rho(P)$) such that $A = sI - P$.

The partial matrix B is a **partial M -matrix** (respectively, **partial M_0 -matrix**) if every fully specified principal submatrix of B is an M -matrix (respectively, M_0 -matrix) and every specified off-diagonal entry of B is nonpositive.

If B is a partial matrix that includes all diagonal entries, the **zero completion** of B , denoted B_0 , is obtained by setting all unspecified (off-diagonal) entries to 0.

Facts:

1. For a Z -matrix A (i.e., every off-diagonal entry of A is nonpositive), the following are equivalent:
 - (a) A is an M -matrix.
 - (b) Every principal minor of A is positive.
 - (c) A is positive stable.
 - (d) A^{-1} is nonnegative.

The analogs of the first three conditions are equivalent to A being an M_0 -matrix. See Section 9.5 for more information about M - and M_0 -matrices.

2. A principal submatrix of an M - (M_0 -) matrix is an M - (M_0 -) matrix (cf. Fact 5 in Section 35.1).
3. [JS96], [Hog01] A partial M - (M_0 -) matrix B that includes all diagonal entries can be completed to an M - (M_0 -) matrix if and only if the zero completion B_0 is an M - (M_0 -) matrix.
4. [Hog98b], [Hog01] A digraph G with a loop at every vertex has the M - (M_0 -) completion property if and only if every strongly connected induced subdigraph of G is a clique.
5. [Hog98b] A digraph G has the M -completion property if and only if the subdigraph induced by the vertices of G that have loops has M -completion.
6. [Hog01] A digraph G has the M_0 -completion property if and only if for every strongly connected induced subdigraph H of G , either H is a clique or H has no loops.
7. [Hog02] Symmetric M - and M_0 -matrices and partial matrices are defined in the obvious way. A graph has the symmetric M -completion property if and only if every connected component of the subgraph induced by the vertices with loops is a clique. A graph has the symmetric M_0 -completion property if and only if every connected component is either a clique or has no loops.

Examples:

The graphs and digraphs for the examples can be found in Examples 1 and 4 of Section 35.1.

1. Even if the digraph of a partial M -matrix does not have the M -matrix completion property, the matrix may still have an M -matrix completion. By Fact 3 it is easy to determine whether there

is an M -completion. For example, complete the partial M -matrix $B = \begin{bmatrix} 1 & ? & ? & -2 \\ -0.5 & 2 & ? & ? \\ ? & -1 & 1 & ? \\ ? & ? & -1 & 1 \end{bmatrix}$

to B_0 by setting every unspecified entry to 0. To determine whether B_0 is an M -matrix, compute the eigenvalues: $\sigma(B_0) = \{2.38028, 1.21945 \pm 0.914474i, 0.180827\}$. If we change the values of entries we can obtain a partial M -matrix that does not have an M -matrix completion,

e.g., $C = \begin{bmatrix} 1 & ? & ? & -2 \\ -2.5 & 2 & ? & ? \\ ? & -1 & 1 & ? \\ ? & ? & -1 & 1 \end{bmatrix}$. Then $\sigma(C_0) = \{2.82397, 1.23609 \pm 1.43499i, -0.296153\}$, so

by Fact 3, C cannot be completed to an M -matrix. Note $\mathcal{D}(B) = \mathcal{D}(C)$ is the digraph 4i, which does not have the M -matrix completion property by Fact 4.

2. The digraphs 4j, 4m, 4p, and 4q have the M - and M_0 -completion properties by Fact 4.
3. The graphs 1a, 1b, 1c, 1d, 1e, 1f, 1g, and 1h and the digraphs 4f, 4g, 4h, 4i, 4k, 4l, 4n, 4o, and 4r have neither the M -completion property nor the M_0 -completion property by Fact 4.
4. The graphs 1j, 1l, and 1m and the digraphs 4a, 4b, 4c, and 4d have the M -completion property by Facts 5 and 4. Of these (di)graphs, only 1l and 4c have the M_0 completion property, by Fact 6.
5. The graphs 1i, 1k, and 1n and the digraph 4e do not have the M -completion property (respectively, the M_0 -completion property) by Facts 5 and 4 (respectively, Fact 6).

35.7 Inverse M -Matrices

In this section, all matrices are real.

Definitions:

The matrix A is an **inverse M -matrix** if A is the inverse of an M -matrix.

The partial matrix B is a **partial inverse M -matrix** if every fully specified principal submatrix of B is an inverse M -matrix and every specified entry of B is nonnegative.

A digraph is **cycle-clique** if the induced subdigraph of every cycle is a clique.

An **alternate path to a single arc** in a digraph G is a path of length greater than 1 between vertices i and j such that the arc (i, j) is in G .

A digraph G is **path-clique** if the induced subdigraph of every alternate path to a single arc is a clique.

A digraph is **homogeneous** if it is either symmetric or asymmetric.

Facts:

1. A matrix A is an inverse M -matrix if and only if all entries of A are nonnegative and all off-diagonal entries of A^{-1} are nonpositive. There are many equivalent characterizations of inverse M -matrices; see Section 9.5 for more information.

2. [JS96] Let $B = \begin{bmatrix} B_{11} & \mathbf{b}_{12} & ? \\ \mathbf{b}_{21}^T & b_{22} & \mathbf{b}_{23}^T \\ ? & \mathbf{b}_{32} & B_{33} \end{bmatrix}$ be an $n \times n$ partial inverse M matrix, where B_{11} and B_{33} are

square matrices of size k and $n - k - 1$, and all entries of the submatrices shown are specified. Then

$$A = \begin{bmatrix} B_{11} & \mathbf{b}_{12} & \mathbf{b}_{12} b_{22}^{-1} \mathbf{b}_{21}^T \\ \mathbf{b}_{21}^T & b_{22} & \mathbf{b}_{23}^T \\ \mathbf{b}_{32} b_{22}^{-1} \mathbf{b}_{21}^T & \mathbf{b}_{32} & B_{33} \end{bmatrix} \text{ is the unique inverse } M\text{-completion of } B \text{ such that } A^{-1} \text{ has}$$

zeros in all the positions where B has unspecified entries. This method can be used to complete a partial inverse M -matrix whose digraph is block-clique.

3. [JS96], [JS99], [Hog98a], [Hog00], [Hog02] A symmetric digraph with a loop at every vertex has the inverse M -completion property if and only if it is block-clique. A digraph obtained from a block-clique digraph by deleting loops from vertices not contained in any block of order greater than 2 also has the inverse M -completion property, and any symmetric digraph that has the inverse M -completion property has that form. The same is true for the symmetric inverse M -completion property (with the obvious definition).
4. [Hog98a] A digraph with a loop at every vertex has the inverse M -completion property if and only if G is path-clique and cycle-clique.
5. [Hog00], [Hog01] A digraph G has the inverse M -completion property if and only if it is path-clique and every strongly connected nonseparable induced subdigraph has the inverse M -completion property. A strongly connected nonseparable digraph is homogeneous. A simple cycle with at least one vertex that does not have a loop has the inverse M -completion property.

Examples:

The graphs and digraphs for the examples can be found in Examples 1 and 4 of Section 35.1.

1. Let $B = \begin{bmatrix} 3 & 1 & ? & ? \\ 4 & 2 & 4 & 2 \\ ? & 1 & 5 & 1 \\ ? & 1 & 2 & 2 \end{bmatrix}$. The completion given by Fact 2 is $A = \begin{bmatrix} 3 & 1 & 2 & 1 \\ 4 & 2 & 4 & 2 \\ 2 & 1 & 5 & 1 \\ 2 & 1 & 2 & 2 \end{bmatrix}$, and

$$A^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} & 0 & 0 \\ -2 & \frac{7}{3} & -\frac{2}{3} & -1 \\ 0 & -\frac{1}{6} & \frac{1}{3} & 0 \\ 0 & -\frac{1}{2} & 0 & 1 \end{bmatrix}.$$

2. By Fact 3, the graphs 1f and 1k have the inverse M -completion property, and these are the only graphs in Example 1 that do.
3. The digraphs 4j and 4m have the inverse M -completion property by Fact 4.
4. The digraphs 4f, 4g, 4h, 4i, 4k, 4l, 4n, 4o, 4p, 4q, and 4r do not have the inverse M -completion property by Fact 4.
5. The digraphs 4a, 4b, 4c, 4d, and 4e do not have the inverse M -completion property by Fact 5.

35.8 P -, $P_{0,1}$ -, and P_0 -Matrices

In this section, all matrices are real.

Definitions:

The matrix A is a P -**matrix** (respectively, P_0 -**matrix**, $P_{0,1}$ -**matrix**) if every principal minor of A is positive (respectively, nonnegative, nonnegative and every diagonal element is positive).

The partial matrix B is a **partial P -matrix** (respectively, **partial P_0 -matrix**, **partial $P_{0,1}$ -matrix**) if every fully specified principal submatrix of B is a P -matrix (respectively, P_0 -matrix, $P_{0,1}$ -matrix).

Facts:

1. A positive definite matrix, M -matrix, or inverse M -matrix is a P -matrix. A positive semidefinite matrix or M_0 -matrix is a P_0 -matrix. See [HJ91] for more information on P -, $P_{0,1}$ -, and P_0 -matrices. A principal submatrix of a P - ($P_{0,1}$ -, P_0 -) matrix is a P - ($P_{0,1}$ -, P_0 -) matrix (cf. Fact 5 in section 35.1).
2. [Hog03a] If a digraph has the P_0 -completion property, then it has the $P_{0,1}$ -completion property. If a digraph has the $P_{0,1}$ -completion property, then it has the P -completion property.
3. [JK96], [Hog01] A digraph has the P -completion property if and only if the subdigraph induced by the vertices that have loops has the P -completion property.
4. [JK96] Every symmetric digraph has the P -completion property. The P -completion of a combinatorially symmetric partial P -matrix can be accomplished by selecting one pair of unspecified entries at a time and choosing the entries of opposite sign and large enough magnitude to make the determinants of all principal matrices completed positive.
5. [JK96] Every order 3 digraph has the P -completion property, but there is an order 4 digraph (see the digraph 4k in Example 4 of Section 35.1) that does not have the P -completion property. [DH00] extended a revised version of this example, digraph 4r, to a family of digraphs, called *minimally chordal symmetric Hamiltonian*, that do not have the P -completion property. The digraph 4n is another example of a digraph in this family (so both 4r and 4n do not have the P -completion property).
6. [DH00] A digraph that can be made symmetric by adding arcs one at a time so that at each stage at most one order 3 induced subdigraph (and no larger) becomes a clique, has the P -completion property.
7. [CDH02] A partial P - ($P_{0,1}$ -, P_0 -) matrix whose digraph is asymmetric can be completed to a P - ($P_{0,1}$ -, P_0 -) matrix as follows: If $i \neq j$ and b_{ji} is specified, then set $x_{ij} = -b_{ji}$. Otherwise, set $x_{ij} = 0$. Every asymmetric digraph has the P - ($P_{0,1}$ -, P_0 -) completion property.

8. [FJT00] Let $B = \begin{bmatrix} B_{11} & \mathbf{b}_{12} & ? \\ \mathbf{b}_{21}^T & b_{22} & \mathbf{b}_{23}^T \\ ? & \mathbf{b}_{32} & B_{33} \end{bmatrix}$ be an $n \times n$ partial P - ($P_{0,1}$ -) matrix, where B_{11} and B_{33} are

square matrices of size k and $n - k - 1$, and all entries of the submatrices shown are specified.

Then $A = \begin{bmatrix} B_{11} & \mathbf{b}_{12} & \mathbf{b}_{12} b_{22}^{-1} \mathbf{b}_{21}^T \\ \mathbf{b}_{21}^T & b_{22} & \mathbf{b}_{23}^T \\ 0 & \mathbf{b}_{32} & B_{33} \end{bmatrix}$ is a P - ($P_{0,1}$ -) completion of B . This method can be used to

complete any partial P - ($P_{0,1}$ -) matrix whose digraph is block-clique. (See [Hog01] for the details of the analogous completion for P_0 -matrices.)

9. [FJT00] Every block-clique digraph has the P - ($P_{0,1}$ -, P_0 -) completion property.
10. [Hog01] A digraph G has the P - (respectively, $P_{0,1}$ -, P_0 -) completion property if and only if every strongly connected and nonseparable induced subdigraph of G has the P - (respectively, $P_{0,1}$ -, P_0 -) completion property. A method to obtain such a completion is given in Algorithm 2.

Algorithm 2:

Let B be a partial P - ($P_{0,1}$ -, P_0 -) matrix such that every strongly connected and nonseparable induced subdigraph K of $\mathcal{D}(B)$ has the P - ($P_{0,1}$ -, P_0 -) property.

1. For each such K , complete the principal submatrix of B corresponding to K to obtain a partial P - ($P_{0,1}$ -, P_0 -) matrix B_1 such that each strongly connected induced subdigraph S of $\mathcal{D}(B_1)$ is block-clique.
 2. For each such S , complete the principal submatrix of B_1 corresponding to S to obtain a partial P - ($P_{0,1}$ -, P_0 -) matrix B_2 .
 3. Set any remaining unspecified entries to 0.
11. [Hog01] A digraph that omits all loops has the P - ($P_{0,1}$ -, P_0 -) completion property. Each connected component of a symmetric digraph that has the P_0 -completion property must have a loop at every vertex or omit all loops.
 12. [Hog01], [CDH02], [JK96] A symmetric n -cycle with a loop at every vertex has the P_0 -completion property if and only if $n \neq 4$. A symmetric n -cycle with a loop at every vertex has the P - and $P_{0,1}$ -completion properties for all n .
 13. [CDH02] All order 2, order 3, and order 4 digraphs with a loop at every vertex have been classified as having or not having the P_0 -completion property. There are some order 4 digraphs that (in 2005) have not been classified as to the P - and $P_{0,1}$ -completion properties.

Examples:

The graphs and digraphs for the examples can be found in Examples 1 and 4 of Section 35.1.

1. It is easy to verify that $B = \begin{bmatrix} 1 & 2 & x_{13} & -1 \\ 1 & 3 & -2 & x_{24} \\ x_{31} & 2 & 1 & 1 \\ -1 & x_{42} & 1 & 2 \end{bmatrix}$ is a partial P -matrix. The graph 1d, called the

double triangle, is interpreted as the digraph of B . Let $x_{24} = y$ and $x_{42} = -y$. A choice of y completes three principal minors, $\det B[\{2, 4\}] = 6 + y^2$, $\det B[\{1, 2, 4\}] = -1 - y + y^2$, and $\det B[\{2, 3, 4\}] = 11 + 4y + y^2$. The choice $y = 2$ makes all three minors positive. Let $x_{24} = z$ and $x_{42} = -z$. With $y = 2$, a choice of z completes four principal minors, $\det B[\{1, 3\}] = 1 + z^2$, $\det B[\{1, 2, 3\}] = 5 + 6z + 3z^2$, $\det B[\{1, 3, 4\}] = 2z^2$, and $\det B = 10 + 18z + 10z^2$, so setting $z = 0$ completes B

to the P -matrix $\begin{bmatrix} 1 & 2 & 0 & -1 \\ 1 & 3 & -2 & 2 \\ 0 & 2 & 1 & 1 \\ -1 & -2 & 1 & 2 \end{bmatrix}$. Any partial P -matrix specifying the double triangle can

be completed in a similar manner, so the double triangle has the P -completion property.

2. The double triangle 1d does not have the P_0 -completion property because $B = \begin{bmatrix} 1 & 2 & 1 & ? \\ -1 & 0 & 0 & -2 \\ -1 & 0 & 0 & -1 \\ ? & 1 & 1 & 1 \end{bmatrix}$

cannot be completed to a P_0 -matrix ([JK96]). This implies the graphs 1g and 1h do not have the

P_0 -completion property by Fact 7 in Section 35.1. The double triangle does have the $P_{0,1}$ -completion property [Wan05].

3. All graphs in Example 1 have the P -completion property by Fact 4.
4. The digraphs 4c and 4d have the P -completion property by Fact 3. Fact 3 can also be applied in conjunction with other facts to several other digraphs in Example 4.
5. The digraphs 4f, 4g, 4h, 4p, and 4q have the P -completion property by Fact 5.
6. The digraphs 4a, 4b, 4c, 4d, 4i, 4j, 4l, 4m, and 4o have the P -completion property by Fact 6.
7. The digraphs 4b, 4h, 4i, 4j, and 4l have the P -, $P_{0,1}$ -, and P_0 -completion properties by Fact 7.

8. The completion of $\begin{bmatrix} 1 & -1 & ? & ? \\ 3 & 2 & -4 & 2 \\ ? & 1 & 5 & -1 \\ ? & 1 & 2 & 2 \end{bmatrix}$ given by Fact 8 is $\begin{bmatrix} 1 & -1 & 2 & -1 \\ 3 & 2 & -4 & 2 \\ 0 & 1 & 5 & -1 \\ 0 & 1 & 2 & 2 \end{bmatrix}$.

9. The graph 1f has the P_0 - and $P_{0,1}$ -completion properties by Fact 9. (It also has the P -completion property but one would not normally cite Fact 9 for that.)
10. The digraphs 4m, 4p, and 4q have the P -, $P_{0,1}$ -, and P_0 -completion properties by Fact 10. Fact 10 can also be applied in conjunction with other facts to several other digraphs in Example 4.
11. The graph 1l and the digraph 4c have the P -, P_0 -, and $P_{0,1}$ -completion properties by Fact 11.
12. The graphs 1i, 1j, 1k, 1m, and 1n do not have the P_0 -completion property by Fact 11.
13. The graphs 1b and 1c have the P_0 - ($P_{0,1}$)-completion property by Fact 12.
14. The graph 1a does not have the P_0 -completion property, but does have the $P_{0,1}$ -completion property, by Fact 12.
15. The graphs 1e and 1g do not have the P_0 -completion property by Fact 12 and by Fact 7 in Section 35.1.

35.9 Positive P -, Nonnegative P -, Nonnegative $P_{0,1}$ -, and Nonnegative P_0 -Matrices

In this section, all matrices are real.

Definitions:

The matrix A is a **positive** (respectively, **nonnegative**) P -**matrix** if A is a P -matrix and every entry of A is positive (respectively, nonnegative). The matrix A is a **nonnegative P_0 -matrix** (respectively, **nonnegative $P_{0,1}$ -matrix**) if A is a P_0 -matrix (respectively, $P_{0,1}$ -matrix) and every entry of A is nonnegative.

The partial matrix B is a **partial positive P -matrix** (respectively, **partial nonnegative P -matrix**, **partial nonnegative P_0 -matrix**, **partial nonnegative $P_{0,1}$ -matrix**) if and only if every fully specified principal submatrix of B is a positive P -matrix (respectively, nonnegative P -matrix, nonnegative P_0 -matrix, partial nonnegative $P_{0,1}$ -matrix) and all specified entries are positive (respectively, nonnegative, nonnegative, nonnegative).

Facts:

1. [Hog03a], [Hog03b] If a digraph has the nonnegative P_0 -completion property, then it has the nonnegative $P_{0,1}$ -completion property. If a digraph has the nonnegative $P_{0,1}$ -completion property, then it has the nonnegative P -completion property. If a digraph has the nonnegative P -completion property, then it has the positive P -completion property.
2. [Hog01] A digraph has the positive (respectively, nonnegative) P -completion property if and only if the subdigraph induced by vertices that have loops has the positive (respectively, nonnegative) P -completion property.

3. [FJT00], [Hog01], [CDH03] All order 2 and order 3 digraphs that have a loop at every vertex have the positive P - (nonnegative P_- , nonnegative $P_{0,-}$, nonnegative $P_{0,1,-}$) completion property.
4. [BEH06] Suppose G is a digraph such that by adding arcs one at a time so that at each stage at most one order 3 induced subdigraph (and no larger) becomes a clique, it is possible to obtain a digraph G' that has the positive P - (respectively, nonnegative P_- , nonnegative $P_{0,-}$, nonnegative $P_{0,1,-}$) completion property. Then G has the positive P - (respectively, nonnegative P_- , nonnegative $P_{0,-}$, nonnegative $P_{0,1,-}$) completion property.
5. [FJT00] A block-clique digraph has the positive P - (nonnegative P_- , nonnegative $P_{0,-}$, nonnegative $P_{0,1,-}$) completion property. See Fact 8 of section 35.8 for information on the construction.
6. [Hog01] A digraph G has the positive P - (respectively, nonnegative P_- , nonnegative $P_{0,-}$, nonnegative $P_{0,1,-}$) completion property if and only if every strongly connected and nonseparable induced subdigraph of G has the positive P - (respectively, nonnegative P_- , nonnegative $P_{0,-}$, nonnegative $P_{0,1,-}$) completion property. See Algorithm 2 of section 35.8 for information on the construction.
7. [Hog01] A digraph that omits all loops has the positive P - (nonnegative P_- , nonnegative $P_{0,1,-}$, nonnegative $P_{0,-}$) completion property. Each connected component of a symmetric digraph that has the nonnegative P_0 -completion property must have a loop at every vertex or omit all loops.
8. [CDH03] An order 4 digraph with a loop at every vertex has the nonnegative P_0 -completion property if and only if it does not contain a 4-cycle or is the clique on 4-vertices. This characterization does not extend to higher order digraphs.
9. [BEH06], [JTU03] All order 4 digraphs that have a loop at every vertex have been classified as to the positive P - (nonnegative P_-) completion property.
10. [CDH03], [FJT00] A symmetric n -cycle that has a loop at every vertex has the nonnegative P_0 -completion property if and only if $n \neq 4$. A symmetric n -cycle that has a loop at every vertex has the positive P - (nonnegative P_- , nonnegative $P_{0,1,-}$) completion property.
11. [BEH06] A minimally chordal symmetric Hamiltonian digraph (cf. Fact 5 in Section 35.8) has neither the positive nor the nonnegative P -completion property.

Examples:

The graphs and digraphs for the examples can be found in Examples 1 and 4 of Section 35.1.

1. The digraphs 4f, 4g, 4h, 4p, and 4q have the positive P - (nonnegative P_- , nonnegative $P_{0,-}$, nonnegative $P_{0,1,-}$) completion property by Fact 3.
2. The graph 1f has the positive P - (nonnegative P_- , nonnegative $P_{0,-}$, nonnegative $P_{0,1,-}$) completion property by Fact 5.
3. The graphs 1j, 1k, 1l, 1m, and 1n and the digraphs 4c and 4d have the positive P - (nonnegative P_-) completion property by Facts 2 and 5.
4. The digraphs 4j, 4m, 4p, and 4q have the positive P - (nonnegative P_- , nonnegative $P_{0,-}$, nonnegative $P_{0,1,-}$) completion property by Fact 6.
5. The graph 1l and the digraph 4c have the positive P_- , nonnegative P_- , nonnegative $P_{0,-}$, and nonnegative $P_{0,1,-}$ -completion properties by Fact 7.
6. The graphs 1i, 1j, 1k, 1m, and 1n do not have the nonnegative P_0 -completion property by Fact 7.
7. The graphs 1a and 1d and the digraphs 4i, 4k, and 4r do not have the nonnegative P_0 -completion property by Fact 8. The graphs 1e, 1g, and 1h and the digraphs 4l, 4n, and 4o do not have the nonnegative P_0 -completion property by Fact 8, using Fact 7 of Section 35.1.
8. By Fact 10, the graph 1a does not have and the graphs 1b and 1c do have the nonnegative P_0 -completion property.
9. The graphs 1a, 1b, and 1c have the positive P - (nonnegative P_- , nonnegative $P_{0,1,-}$) completion property by Fact 10.

35.10 Entry Sign Symmetric P -, Entry Sign Symmetric P_0 -, Entry Sign Symmetric $P_{0,1}$ -, Entry Weakly Sign Symmetric P -, and Entry Weakly Sign Symmetric P_0 -Matrices

In this section, all matrices are real. In the literature, *entry sign symmetric* is often called *sign symmetric*; as defined in Chapter 19.2, the latter term is used for a different condition.

Definitions:

The matrix A is an **entry sign symmetric P -** (respectively, $P_{0,1}$ -, P_0 -) **matrix** if and only if A is a P -matrix (respectively, $P_{0,1}$ -matrix, P_0 -matrix) and for all i, j , either $a_{ij}a_{ji} > 0$ or $a_{ij} = a_{ji} = 0$.

The matrix A is an **entry weakly sign symmetric P -** (respectively, $P_{0,1}$ -, P_0 -) **matrix** if and only if A is a P -matrix (respectively, $P_{0,1}$ -matrix, P_0 -matrix) and for all i, j , $a_{ij}a_{ji} \geq 0$.

The partial matrix B is a **partial entry sign symmetric P -** (respectively, $P_{0,1}$ -, P_0 -) **matrix** if and only if every fully specified principal submatrix of B is an entry sign symmetric P - (respectively, $P_{0,1}$ -, P_0 -) matrix and if both b_{ij} and b_{ji} are specified then $b_{ij}b_{ji} > 0$ or $b_{ij} = b_{ji} = 0$.

The partial matrix B is a **partial entry weakly sign symmetric P -** (respectively, $P_{0,1}$ -, P_0 -) **matrix** if and only if every fully specified principal submatrix of B is an entry weakly sign symmetric P - (respectively, $P_{0,1}$ -, P_0 -) matrix and if both b_{ij} and b_{ji} are specified then $b_{ij}b_{ji} \geq 0$.

Facts:

1. [Hog03a] Any pattern that has the entry sign symmetric P_0 - (respectively, entry weakly sign symmetric P_0 -) completion property also has the entry sign symmetric $P_{0,1}$ - (respectively, entry weakly sign symmetric $P_{0,1}$ -) completion property. Any pattern that has the entry sign symmetric $P_{0,1}$ - (respectively, entry weakly sign symmetric $P_{0,1}$ -) completion property also has the entry sign symmetric P - (respectively, entry weakly sign symmetric P -) completion property.
2. [Hog01] A digraph G has the (weakly) entry sign symmetric P -completion property if and only if the subdigraph of G induced by vertices that have loops has the entry (weakly) sign symmetric P -completion property.
3. [FJT00], [Hog01] A digraph G has the entry sign symmetric P_0 -completion property if and only if for every connected component H of G , either H omits all loops or H has a loop at every vertex and is block-clique.
4. [FJT00] A symmetric digraph with a loop at every vertex has the entry sign symmetric $P_{0,1}$ -completion property if and only if every connected component is block-clique.
5. [FJT00] A block-clique digraph has the entry sign symmetric P - (entry sign symmetric $P_{0,1}$ -, entry sign symmetric P_0 -, entry weakly sign symmetric P -, entry weakly sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion property. (See Fact 8 of Section 35.8 for information on the construction.)
6. [Hog01] A digraph G has the entry sign symmetric P - (entry weakly sign symmetric P -, entry weakly sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion property if and only if every strongly connected and nonseparable induced subdigraph of G has the entry sign symmetric P - (entry weakly sign symmetric P -, entry weakly sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion property. (See Algorithm 2 of Section 35.8 for information on the construction.)
7. Fact 6 is not true for the entry sign symmetric P_0 -matrices (cf. Fact 3) or for the entry sign symmetric $P_{0,1}$ -matrices [Wan05]. In particular, the digraphs 4p and 4q in Example 4 of Section 35.1 have neither the entry sign symmetric P_0 -completion property nor the entry sign symmetric $P_{0,1}$ -completion property.
8. [DHH03] A symmetric n -cycle with a loop at every vertex has the entry sign symmetric P - (entry weakly sign symmetric P -, entry weakly sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion property if and only if $n \neq 4$ and $n \neq 5$.

9. [DHH03] An order 3 digraph G with a loop at every vertex has the entry sign symmetric P - (entry weakly sign symmetric P -, entry weakly sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion property if and only if its digraph does not contain a 3-cycle or is a clique.
10. [DHH03], [Wan05] All order 4 digraphs that have a loop at every vertex have been classified as to the entry sign symmetric P - (entry sign symmetric $P_{0,1}$ -, entry sign symmetric P_0 -, entry weakly sign symmetric P -, entry weakly sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion property.

Examples:

The graphs and digraphs for the examples can be found in Examples 1 and 4 of section 35.1.

1. By Fact 3, the graphs 1f and 1l and the digraph 4c have the entry sign symmetric P_0 -completion property and none of the other graphs or digraphs pictured do.
2. The graphs 1a, 1b, 1c, 1d, 1e, 1g, and 1h do not have the entry sign symmetric $P_{0,1}$ -completion proper by Fact 4.
3. The graph 1f has the entry sign symmetric P - (entry sign symmetric $P_{0,1}$ -, entry sign symmetric P_0 -, entry weakly sign symmetric P -, entry sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion proper by Fact 5.
4. The graphs 1j, 1k, 1l, 1m, and 1n and the digraphs 4c and 4d and have the entry (weakly) sign symmetric P -completion property by Facts 2 and 5.
5. The digraphs 4j, 4m, 4p, and 4q have the entry sign symmetric P - (entry weakly sign symmetric P -, entry weakly sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion property by Fact 6.
6. The graphs 1a and 1b do not have the entry sign symmetric P - (entry weakly sign symmetric P -, entry weakly sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion property by Fact 8.
7. The graph 1c has the entry sign symmetric P - (entry weakly sign symmetric P -, entry weakly sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion property by Fact 8.
8. The digraphs 4f, 4g, 4h, 4k, 4l, 4n, 4o, and 4r do not have the entry sign symmetric P - (entry weakly sign symmetric P -, entry weakly sign symmetric $P_{0,1}$ -, entry weakly sign symmetric P_0 -) completion property by Fact 9 and Fact 7 in section 35.1.

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36

Algebraic Connectivity

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36.1	Algebraic Connectivity for Simple Graphs: Basic Theory	36-1
36.2	Algebraic Connectivity for Simple Graphs: Further Results	36-3
36.3	Algebraic Connectivity for Trees	36-4
36.4	Fiedler Vectors and Algebraic Connectivity for Weighted Graphs	36-7
36.5	Absolute Algebraic Connectivity for Simple Graphs	36-9
36.6	Generalized Laplacians and Multiplicity	36-10
	References	36-11

36.1 Algebraic Connectivity for Simple Graphs: Basic Theory

Let G be a simple graph on $n \geq 2$ vertices with Laplacian matrix L_G , and label the eigenvalues of L_G as $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n$. Throughout this chapter, we consider only Laplacian matrices for graphs on at least two vertices. Henceforth, we use the term *graph* to refer to a simple graph.

Definitions:

The **algebraic connectivity** of G , denoted $\alpha(G)$, is given by $\alpha(G) = \mu_2$.

A **Fiedler vector** is an eigenvector of L_G corresponding to $\alpha(G)$.

Given graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, their **product**, $G_1 \times G_2$, is the graph with vertex set $V_1 \times V_2$, with vertices (u_1, u_2) and (w_1, w_2) adjacent if and only if either u_1 is adjacent to w_1 in G_1 and $u_2 = w_2$, or u_2 is adjacent to w_2 in G_2 and $u_1 = w_1$.

Facts:

- [Fie89] Let G be a graph of order n with Laplacian matrix L_G . Then $\alpha(G) = \min\{\sum_{i < j, \{i,j\} \in E} (x_i - x_j)^2 \mid \sum_{1 \leq i \leq n} x_i^2 = 1, \sum_{1 \leq i \leq n} x_i = 0\} = \min\{x^T L_G x \mid x^T x = 1, x^T \mathbf{1} = 0\}$.
- [Fie73] The algebraic connectivity of a graph is nonnegative, and is equal to 0 if and only if the graph is disconnected.
- [Fie73] Let G be a connected graph on n vertices with vertex connectivity $\kappa_v(G)$, and suppose that $G \neq K_n$. Then $\alpha(G) \leq \kappa_v(G)$. (See Fact 13 below for a discussion of the equality case.)
- [Fie73] Suppose that G is a graph on n vertices, and let $\overline{G}$ denote the complement of G . Then $\alpha(\overline{G}) = n - \mu_n$, where μ_n denotes the largest Laplacian eigenvalue for G .

5. ([GR01], p. 280) If G is a graph on $n \geq 2$ vertices that is regular of degree k , then denoting the eigenvalues of $\mathcal{A}_G$ by $\lambda_1 \leq \dots \leq \lambda_n$, we have $\alpha(G) = k - \lambda_{n-1}$.
6. [Mer94] Suppose that G_1 and G_2 are two graphs on n_1 and n_2 vertices, respectively, with $n_1, n_2 \geq 2$. Then $\alpha(G_1 + G_2) = \min\{\alpha(G_1) + n_2, \alpha(G_2) + n_1\}$. Similarly, if H is a graph on $k \geq 2$ vertices, then $\alpha(H + K_1) = \alpha(H) + 1$.
7. [Fie73] Suppose that G_1 and G_2 are graphs, each of which has at least two vertices. Then $\alpha(G_1 \times G_2) = \min\{\alpha(G_1), \alpha(G_2)\}$.
8. [Fie73] Suppose that the graph $\hat{G}$ is formed from the graph G by adding an edge not already present in G . Then $\alpha(G) \leq \alpha(\hat{G})$.
9. [FK98] Let G and H be graphs, and suppose that the graph $\hat{G}$ is formed from $G \cup H$ as follows: Fix a vertex v of G and a subset S of the vertex set for H , and for each $w \in S$, add in the edge between v and w . Then $\alpha(\hat{G}) \leq \alpha(G)$.
10. [Fie73] Let G be a graph, and suppose that the graph $\hat{G}$ is formed from G by deleting a collection of k vertices and all edges incident with them. Then $\alpha(\hat{G}) \geq \alpha(G) - k$.
11. [GMS90] Let G be a graph on $n \geq 3$ vertices and suppose that the edge e of G is not on any 3-cycles. Form $\hat{G}$ from G by deleting e and identifying the two vertices incident with it. Then $\alpha(\hat{G}) \geq \alpha(G)$.
12. [Fie73] If G is a graph on n vertices, then $\alpha(G) \leq n$. Equality holds if and only if $G = K_n$.
13. [KMNS02] Let G be a connected graph on n vertices with vertex connectivity $\kappa_v(G)$, and suppose that $G \neq K_n$. Then $\alpha(G) = \kappa_v(G)$ if and only if G can be written as $G = G_1 + G_2$, where G_1 is disconnected, G_2 has $\kappa_v(G)$ vertices, and $\alpha(G_2) \geq 2\kappa_v(G) - n$.
14. [Fie89] If G is a connected graph on n vertices with edge connectivity $\kappa_e(G)$, then $2(1 - \cos(\frac{\pi}{n}))\kappa_e(G) \leq \alpha(G)$. Equality holds if and only if $G = P_n$.

Examples:

1. ([GK69], p. 138) For $n \geq 2$, the algebraic connectivity of the path P_n is $2(1 - \cos(\frac{\pi}{n}))$, and it is a simple eigenvalue of the corresponding Laplacian matrix.
2. The following can be deduced from basic results on circulant matrices. If $n \geq 3$, the algebraic connectivity of the cycle C_n is $2(1 - \cos(\frac{2\pi}{n}))$, and it is an eigenvalue of multiplicity 2 of the corresponding Laplacian matrix.
3. The algebraic connectivity of K_n is n , and it is an eigenvalue of multiplicity $n - 1$ of the corresponding Laplacian matrix.
4. If $m \leq n$ and $2 \leq n$, then $\alpha(K_{m,n}) = m$. If $1 \leq m < n$, then m is an eigenvalue of multiplicity $n - 1$ of the corresponding Laplacian matrix, while if $2 \leq m = n$, then m is an eigenvalue of multiplicity $2m - 2$ of the corresponding Laplacian matrix.
5. The algebraic connectivity of the Petersen graph is 2, and it is an eigenvalue of multiplicity 5 of the corresponding Laplacian matrix.
6. The algebraic connectivity of the ladder on 6 vertices, shown in Figure 36.1, is 1.
7. Graphs with large algebraic connectivity arise in the study of the class of so-called expander graphs. (See Section 28.5 or [Alo86] for definitions and discussion.)
8. A Fiedler vector for a graph provides a heuristic for partitioning its vertex set so that the number of edges between the two parts is small (see [Moh92] for a discussion), and that heuristic has applications to sparse matrix computations (see [PSL90]).

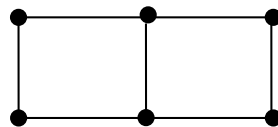


FIGURE 36.1

36.2 Algebraic Connectivity for Simple Graphs: Further Results

The term *graph* means *simple graph* in this section.

Definitions:

Let $G = (V, E)$ be a graph, and let $X, V \setminus X$ be a nontrivial partitioning of V . The corresponding **edge cut** is the set E_X of edges of G that have one end point in X and the other end point in $V \setminus X$.

Suppose that G_1 and G_2 are graphs. A graph H is formed by **appending** G_2 **at vertex** ν of G_1 if H is constructed from $G_1 \cup G_2$ by adding an edge between ν and a vertex of G_2 .

Suppose that $g, n \in \mathbb{N}$ with $n > g \geq 3$. The graph $C_{n,g}$ is formed by appending the cycle C_g at a pendent vertex of the path P_{n-g} .

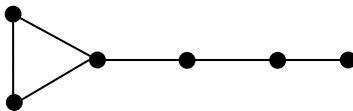
Suppose that $g, n \in \mathbb{N}$ with $n > g \geq 3$. Let $D_{g,n-g}$ denote the graph formed from the cycle C_g by appending $n - g$ isolated vertices at a single vertex of that cycle.

For a connected graph G , a vertex ν is a **cut-vertex** if $G - \nu$, the graph formed from G by deleting the vertex ν and all edges incident with it, is disconnected.

A graph is **unicyclic** if it contains precisely one cycle.

Facts:

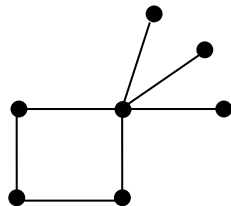
1. [Moh91] If G is a connected graph on n vertices with diameter d , then $\alpha(G) \geq \frac{4}{dn}$.
2. [AM85] If G is a connected graph on n vertices with diameter d and maximum degree Δ , then $d \leq 2 \lceil \sqrt{\frac{2\Delta}{\alpha(G)}} \log_2 n \rceil$.
3. [Moh91] If G is a connected graph on n vertices with diameter d and maximum degree Δ , then $d \leq 2 \lceil \frac{\Delta + \alpha(G)}{4\alpha(G)} \ln(n-1) \rceil$.
4. [Moh92] Let $G = (V, E)$ be a graph, let $X, V \setminus X$ be a nontrivial partitioning of its vertex set, and let E_X denote the corresponding edge cut. Then $\alpha(G) \leq \frac{|V||E_X|}{|X||V \setminus X|}$ and ([FKP03a]) if $\alpha(G) = \frac{|V||E_X|}{|X||V \setminus X|}$, then necessarily there are integers d_1, d_2 such that:
 - Each vertex in X is adjacent to precisely d_1 vertices in $V \setminus X$.
 - Each vertex in $V \setminus X$ is adjacent to precisely d_2 vertices in X .
 - $|X|d_1 = |V \setminus X|d_2$.
 - $\alpha(G) = d_1 + d_2$.
5. [Moh89] Let $G = (V, E)$ be a graph on at least four vertices, with maximum degree Δ . Then $\frac{\alpha(G)}{2} \leq \Phi(G) \leq \sqrt{\alpha(G)(2\Delta - \alpha(G))}$, where $\Phi(G)$ is the isoperimetric number of G . (See Section 32.5.)
6. [FK98] Among all connected graphs on n vertices with girth 3, the algebraic connectivity is uniquely minimized by $C_{n,3}$.
7. [FKP02] If $g \geq 4$ and $n \geq 3g - 1$, then among all connected graphs on n vertices with girth g , the algebraic connectivity is uniquely minimized by $C_{n,g}$.
8. [Kir00] Let G be a graph on n vertices, and suppose that G has k cut-vertices, where $2 \leq k \leq \frac{n}{2}$. Then $\alpha(G) \leq \frac{2(n-k)}{n-k+2+\sqrt{(n-k)^2+4}}$. For each such k and n , there is a graph on n vertices with k cut-vertices such that equality is attained in the upper bound on α .
9. [Kir01] Suppose that $n \geq 5$ and $\frac{n}{2} < k$, and let G be a graph on n vertices with k cut-vertices. Let $q = \lfloor \frac{k}{n-k} \rfloor$ and $l = k - (n-k)q$.
 - If $l = 1$, then $\alpha(G) \leq 2(1 - \cos(\frac{\pi}{2q+3}))$.
 - If $l = 0$, let θ_0 be the unique element of $[\frac{\pi}{2q+3}, \frac{\pi}{2q+1}]$ such that $(n-k-1)\cos((2q+1)\theta_0/2) + \cos((2q+3)\theta_0/2) = 0$. Then $\alpha(G) \leq 2(1 - \cos(\theta_0))$.

FIGURE 36.2 The graph $C_{6,3}$.

- If $2 \leq l$, let θ_0 be the unique element of $\left[-\frac{\pi}{2q+5}, \frac{\pi}{2q+3}\right]$ such that $(n-k-1)\cos((2q+3)\theta_0/2) + \cos((2q+5)\theta_0/2) = 0$. Then $\alpha(G) \leq 2(1 - \cos(\theta_0))$.

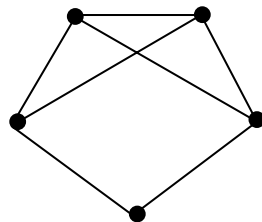
For each k and n with $\frac{n}{2} < k$, there is a graph on n vertices with k cut-vertices such that equality is attained in the corresponding upper bound on α .

- [FK98] Over the class of unicyclic graphs on n vertices having girth 3, the algebraic connectivity is uniquely maximized by $D_{3,n-3}$.
- [FKP03b] Over the class of unicyclic graphs on n vertices having girth 4, the algebraic connectivity is uniquely maximized by $D_{4,n-4}$.
- [FKP03b] Fix $g \geq 5$. There is an N such that if $n > N$, then over the class of unicyclic graphs on n vertices having girth g , the algebraic connectivity is uniquely maximized by $D_{g,n-g}$.
- [Kir03] For each real number $r \geq 0$, there is a sequence of graphs G_k with distinct algebraic connectivities, such that $\alpha(G_k)$ converges monotonically to r as $k \rightarrow \infty$.

FIGURE 36.3 The graph $D_{4,3}$.

Examples:

- The graph $C_{6,3}$ is shown in Figure 36.2; its algebraic connectivity is approximately 0.3249.
- The graph $D_{4,3}$ is shown in Figure 36.3.
- [FK98] The algebraic connectivity of $D_{3,n-3}$ is 1, and it is an eigenvalue of multiplicity $n-3$ of the corresponding Laplacian matrix.
- Consider the graph G pictured in Figure 36.4. Its algebraic connectivity is 2, so that from Fact 5 above, we deduce that $1 \leq \Phi(G) \leq 2\sqrt{2}$. It turns out that the value of $\Phi(G)$ is $\frac{3}{2}$.

FIGURE 36.4 The graph G for Example 4

36.3 Algebraic Connectivity for Trees

The term *graph* means *simple graph* in this section.

Definitions:

Let T be a tree, and let $\mathbf{y}$ be a Fiedler vector for T . A **characteristic vertex** of T is a vertex i satisfying one of the following conditions:

- $y_i = 0$, and vertex i is adjacent to a vertex j such that $y_j \neq 0$.
- There is a vertex j adjacent to vertex i such that $y_i y_j < 0$.

A tree is called **type I** if it has a single characteristic vertex, and **type II** if it has two characteristic vertices.

Let T be a tree and suppose that v is a vertex of T . The **bottleneck matrix** of a branch of T at v is the inverse of the principal submatrix of the Laplacian matrix corresponding to the vertices of that branch.

A branch at vertex v is called a **Perron branch at v** if the Perron value of the corresponding bottleneck matrix is maximal among all branches at v .

Suppose that T is a type I tree with characteristic vertex i . A branch at i is called an **active branch** if, for some Fiedler vector $\mathbf{y}$, the entries in $\mathbf{y}$ corresponding to the vertices in the branch are nonzero.

Suppose that $n \geq d + 1$. Denote by $P(n, d)$ the tree constructed as follows: Begin with the path P_{d+1} , on vertices $1, \dots, d + 1$, labeled so that vertices 1 and $d + 1$ are pendent, while for each $i = 2, \dots, d$, vertex i is adjacent to vertices $i - 1$ and $i + 1$, then append $n - d - 1$ isolated vertices at vertex $\lfloor \frac{d}{2} \rfloor + 1$ of that path.

Let $T(k, l, d)$ be the tree on $n = d + k + l$ vertices constructed from the path P_d by appending k isolated vertices at one end vertex of P_d , and appending l isolated vertices at the other end vertex of P_d .

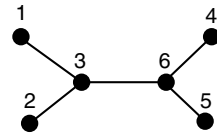
Facts:

1. The bottleneck matrix for a branch is the inverse of an irreducible M -matrix, and so is entrywise positive (see Chapter 9.5 and/or [KNS96]).
2. [Fie89] If T is a tree on n vertices, then $2(1 - \cos(\frac{\pi}{n})) \leq \alpha(T)$; equality holds if and only if $G = P_n$.
3. [Mer87] If T is a tree on $n \geq 3$ vertices, then $\alpha(G) \leq 1$; equality holds if and only if $G = K_{1, n-1}$.
4. [GMS90] If T is a tree with diameter d , then $\alpha(T) \leq 2(1 - \cos(\frac{\pi}{d+1}))$.
5. [Fie75b] Let T be a tree on vertices labeled $1, \dots, n$, and suppose that $\mathbf{y}$ is a Fiedler vector of T . Then exactly one of two cases can occur.
 - There are no zero entries in $\mathbf{y}$. Then T contains a unique edge $\{i, j\}$ such that $y_j < 0 < y_i$. As we move along any path in T that starts at i and does not contain j , the corresponding entries in $\mathbf{y}$ are positive and increasing. As we move along any path in T that starts at j and does not contain i , the corresponding entries in $\mathbf{y}$ are negative and decreasing. In this case T is type II.
 - The vector $\mathbf{y}$ has at least one zero entry. Then there is a unique vertex i of T such that $y_i = 0$ and i is adjacent to a vertex j such that $y_j \neq 0$. As we move along any path in T that starts at vertex i , the corresponding entries in $\mathbf{y}$ are either increasing, decreasing, or identically zero. In this case T is type I.
6. [KNS96] Let T be a tree and $\mathbf{y}$ be a Fiedler vector for T . Let P be a path in T that starts at a characteristic vertex i of T , and does not contain any other characteristic vertices of T . If, as we move along P away from i the entries of $\mathbf{y}$ are increasing, then they are also concave down. If, as we move along P away from i the entries of $\mathbf{y}$ are decreasing, then they are also concave up.
7. [Mer87] Let T be a tree. Then each Fiedler vector for T identifies the same vertex (or vertices) as the characteristic vertex (or vertices). Consequently, the type of the tree is independent of the particular choice of the Fiedler vector.
8. [Fie75a] If T is a type II tree, then $\alpha(T)$ is a simple eigenvalue.
9. [KNS96] A tree T is type I if and only if at some vertex i , there are two or more Perron branches. In that case, i is the characteristic vertex of T , $\alpha(T)$ is the reciprocal of the Perron value of the bottleneck matrix of a Perron branch at i , and ([KF98]) the multiplicity of $\alpha(T)$ is one less than the number of Perron branches at i .
10. [KNS96] A tree T is type II if and only if there is a unique Perron branch at every vertex. In this case, the characteristic vertices of T are the unique adjacent vertices i, j such that the Perron branch at vertex i is the branch containing vertex j , and the Perron branch at vertex j is the branch containing vertex i . Letting B_i and B_j denote the bottleneck matrices for the Perron branches at vertices i and j , respectively, $\exists! \gamma \in (0, 1)$ such that $\rho(B_i - \gamma J) = \rho(B_j - (1 - \gamma)J)$, and the common Perron value for these matrices is $1/\alpha(T)$.
11. The following is a consequence of results in [GM87] and [KNS96]. Suppose that T is a type I tree with characteristic vertex i ; then a branch at i is an active branch if and only if it is a Perron branch at i .
12. [FK98] Among all trees on n vertices with diameter d , the algebraic connectivity is maximized by $P(n, d)$.
13. [FK98] Let T be a tree on n vertices with diameter d . Then $\alpha(T) \geq \alpha(T(\lfloor \frac{n-d+1}{2} \rfloor, \lfloor \frac{n-d+1}{2} \rfloor, d-1))$, with equality holding if and only if $T = T(\lfloor \frac{n-d+1}{2} \rfloor, \lfloor \frac{n-d+1}{2} \rfloor, d-1)$.

Examples:

- [GM90] The algebraic connectivity of $T(k, l, 2)$ is the smallest root of the polynomial $x^3 - (k + l + 4)x^2 + (2k + 2l + kl + 5)x - (k + l + 2)$.
- Let T be the tree constructed as follows: At a single vertex v , append $k \geq 2$ copies of the path P_2 and $l \geq 0$ pendent vertices. For each branch at v consisting of a path on two vertices, the bottleneck matrix can be written as $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, which has Perron value $\frac{3+\sqrt{5}}{2}$, while each branch at v consisting of a single vertex has bottleneck matrix equal to $[1]$. Then $\alpha(T) = \frac{3-\sqrt{5}}{2}$, and is an eigenvalue of multiplicity $k - 1$ of the corresponding Laplacian matrix. In particular, T is a type I tree with characteristic vertex v .
- [FK98] If $d \geq 4$ is even, $\alpha(P(n, d)) = 2(1 - \cos(\frac{\pi}{d+1}))$, and is a simple eigenvalue of the corresponding Laplacian matrix.

- Consider the tree $T(2, 2, 2)$ shown in Figure 36.5. At both of its nonpendent vertices, the bottleneck matrix for the corre-

**FIGURE 36.5** The tree $T(2, 2, 2)$.

sponding Perron component can be written as $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.

It follows that $1/\alpha(T) = \rho \left(\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix} - 1/2J \right) = (5 +$

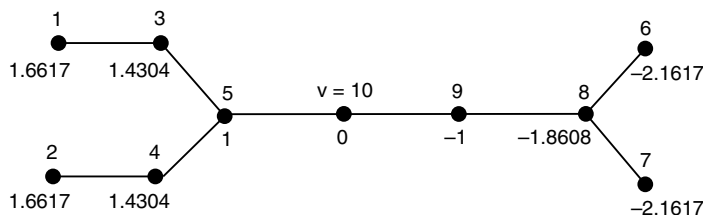
$\sqrt{17})/4$. Hence, the algebraic connectivity for $T(2, 2, 2)$ is $(5 - \sqrt{17})/2$, and it is a type II tree, with the two nonpendent

vertices as the characteristic vertices. $[\frac{3+\sqrt{17}}{4}, \frac{3+\sqrt{17}}{4}, 1, -\frac{3+\sqrt{17}}{4}, -\frac{3+\sqrt{17}}{4}, -1]^T$ is a Fiedler vector.

- [GM87] Consider the tree T shown in Figure 36.6; this is a type I tree with characteristic vertex v . The numbers above the vertices are the vertex numbers and the numbers below are (to four decimal places) the entries in a Fiedler vector for T . The bottleneck matrix for the branch at v on 5 vertices

can be written as $\begin{bmatrix} 3 & 1 & 2 & 1 & 1 \\ 1 & 3 & 1 & 2 & 1 \\ 2 & 1 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$, while that for the branch at v on 4 vertices can be written

$\begin{bmatrix} 3 & 2 & 2 & 1 \\ 2 & 3 & 2 & 1 \\ 2 & 2 & 2 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$. The algebraic connectivity is the smallest root of the polynomial $x^3 - 6x^2 + 8x - 1$, and is approximately 0.1392.

**FIGURE 36.6** The tree T in Example 5.

36.4 Fiedler Vectors and Algebraic Connectivity for Weighted Graphs

The term *graph* means *simple graph* in this section.

Definitions:

Let $G = (V, E)$ be a graph on $n \geq 2$ vertices, and suppose that for each edge $e \in E$ we have an associated positive number $w(e)$.

- Then $w(e)$ is the **weight** of e .
- The function $w : E \rightarrow \mathbb{R}^+$ is a **weight function** on G .
- The graph G , together with the function $w : E \rightarrow \mathbb{R}^+$, is a **weighted graph**, and is denoted G_w .

The **Laplacian matrix** $L(G_w)$ for the weighted graph G_w is the $n \times n$ matrix $L(G_w) = [l_{i,j}]$ such that $l_{i,j} = 0$ if vertices i and j are distinct and not adjacent in G , $l_{i,j} = -w(e)$ if $e = \{i, j\} \in E$, and $l_{i,i} = \sum w(e)$, where the sum is taken over all edges e incident with vertex i .

Throughout this chapter, we only consider Laplacian matrices for weighted graphs on at least two vertices. The notation $L(G_w)$ for the Laplacian matrix of the weighted graph G_w should not be confused with the notation for the line graph introduced in Section 28.2.

Let G_w be a weighted graph on n vertices, and denote the eigenvalues of $L(G_w)$ by $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n$. The **algebraic connectivity** of G_w is μ_2 and is denoted $\alpha(G_w)$.

Let G_w be a weighted graph. A **Fiedler vector** for G_w is an eigenvector of $L(G_w)$ corresponding to the eigenvalue $\alpha(G_w)$.

Let G_w be a weighted graph and $\mathbf{y}$ be a Fiedler vector for G_w . For each vertex i of G_w , the **valuation of i** is equal to the corresponding entry in $\mathbf{y}$. A vertex is **valuated positively, negatively, or zero** accordingly, as the corresponding entry in $\mathbf{y}$ is positive, negative, or zero.

Facts:

1. [Fie75b] Let G_w be a weighted graph and let $\mathbf{y}$ be a Fiedler vector for G_w . Then exactly one of the following holds.
 - Case A—There is a single block B_0 in G_w containing both positively valuated vertices and negatively valuated vertices. For every other block of G_w , the vertices are either all valuated positively or all valuated negatively or all valuated zero. Along any path in G that starts at a vertex $k \in B_0$ and contains at most two cut-vertices from each block, the valuations of the cut-vertices form either an increasing sequence, a decreasing sequence, or an all zero sequence, according as $y_k > 0$, $y_k < 0$, or $y_k = 0$, respectively. In the case where $y_k = 0$, then each vertex on that path is valuated zero.
 - Case B — No block in G_w has both positively and negatively valuated vertices. Then there is a unique vertex i of G_w having zero valuation that is adjacent to a vertex j with nonzero valuation. The vertex i is a cut-vertex. Each block of G_w contains (with the exception of i) only vertices of positive valuation, or only vertices of negative valuation, or only vertices of zero valuation. Along any path that starts at vertex i and contains at most two cut-vertices from each block, the valuations of the cut-vertices form either an increasing sequence, a decreasing sequence, or an all zero sequence. Any path in G_w that contains both positively and negatively valuated vertices must pass through vertex i .
2. [KF98] Let G_w be a weighted graph. If case A holds for some Fiedler vector, then case A holds for every Fiedler vector and, moreover, every Fiedler vector identifies the same block of G_w having vertices of both positive and negative valuations. Similarly, if case B holds for some Fiedler vector, then case B holds for every Fiedler vector and, moreover, every vector identifies the same vertex i that is valuated zero and is adjacent to a vertex with nonzero valuation.

3. [KF98] Suppose that for a weighted graph G_w , case A holds, and let B_0 be the unique block of G_w containing both positively and negatively valuated vertices. Fix an edge $e \in B_0$, and let $\hat{w}$ be the weighting of G formed from w by replacing $w(e)$ by t , where $0 < t < w(e)$. Then for the weighted graph $G_{\hat{w}}$, case A also holds, and B_0 is still the block of G containing both positively and negatively valuated vertices. Similarly, if $B_0 - e$ is a block of $G - e$, then case A still holds for the weighting $\hat{w}$ of $G - e$ arising from setting $\hat{w}(f) = w(f)$ for each $f \in E \setminus \{e\}$, and $B_0 - e$ is still the block containing both positively and negatively valuated vertices.
4. [KF98] Suppose that G is a graph, that w is weight function on G , and consider the weighted graph G_w . Suppose that B is a block of G_w that is valuated all nonnegatively or all nonpositively by some Fiedler vector. Let $\hat{G}_w$ be the weighted graph formed from G_w by either adding a weighted edge to B , or raising the weight of an existing edge in B , and denote the modified block by $\hat{B}$. Then every Fiedler vector for $\hat{G}_w$ valuates the vertices of $\hat{B}$ all nonnegatively, or all nonpositively.
5. [FN02] Suppose that G_w is a weighted graph on n vertices with Laplacian matrix $L(G_w)$. For each nonempty subset U of the vertex set V , let $L(G_w)[U]$ denote the principal submatrix of $L(G_w)$ on the rows and columns corresponding to the vertices in U , and let $\tau(L(G_w)[U])$ denote its smallest eigenvalue. Then $\alpha(G_w) \geq \min\{\max\{\tau(L(G_w)[U]), \tau(L(G_w)[V \setminus U])\} | U \subset V, 0 < |U| \leq \lfloor \frac{n}{2} \rfloor\}$.
6. [KN04] Suppose that $G = (V, E)$ is a graph, that w is weight function on G , and consider the weighted graph G_w . Fix an edge $e = \{i, j\} \in E$. For each $t \geq w(e)$, let w_t be the weight function on G constructed from w by setting the weight of the edge e to t .
 - If some Fiedler vector $\mathbf{y}$ for G_w has the property that $y_i = y_j$, then for any $t \geq w(e)$, $\alpha(G_{w_t}) = \alpha(G_w)$. In particular, if $\alpha(G_w)$ is an eigenvalue of multiplicity at least two, then $\alpha(G_{w_t}) = \alpha(G_w)$ for all $t \geq w(e)$.
 - If $\alpha(G_w)$ is a simple eigenvalue, then $\exists t_0 > 0$ such that for each $t \in [0, t_0]$, $\alpha(G_{w_t})$ is a twice differentiable function of t , with $0 \leq \frac{d\alpha(G_{w_t})}{dt} \leq 2$ and $\frac{d^2\alpha(G_{w_t})}{dt^2} \leq 0$. Further, let $\mathbf{y}(t)$ denote a Fiedler vector for G_{w_t} of norm 1 that is analytic for $t \in [0, t_0]$; then $|(\mathbf{y}(t))_i - (\mathbf{y}(t))_j|$ is nonincreasing for $t \in [0, t_0]$.

Examples:

1. Consider the weighting of K_n in which each edge has weight 1. Then any vector that is orthogonal to $\mathbf{1}_n$ is a Fiedler vector.
2. Consider the weighting of $K_{1,n-1}$ in which edge has weight 1. If i and j are pendent vertices of $K_{1,n-1}$, then $\mathbf{e}_i - \mathbf{e}_j$ is a Fiedler vector.
3. Consider the weighting of the path P_n in which each edge has weight 1, and label the vertices of P_n so that vertices 1 and n are pendent, while for each $i = 2, \dots, n-1$, vertex i is adjacent to vertices $i-1$ and $i+1$. Then the vector $\mathbf{y}$ with $y_i = \cos((2i-1)\pi/(2n))$ is a Fiedler vector.
4. Suppose that $a, b > 0$, and let T_w be the weighting of the path P_4 arising by assigning the pendent edges weight a , and the nonpendent edge weight b . The corresponding Laplacian matrix can be

written as $\begin{bmatrix} a & 0 & -a & 0 \\ 0 & a & 0 & -a \\ -a & 0 & a+b & -b \\ 0 & -a & -b & a+b \end{bmatrix}$, which has eigenvalues $0, a+b \pm \sqrt{a^2+b^2}$, and $2a$. In

particular, $\alpha(T_w) = a+b - \sqrt{a^2+b^2}$, with corresponding Fiedler vector $\begin{bmatrix} a \\ -a \\ \sqrt{a^2+b^2} - b \\ b - \sqrt{a^2+b^2} \end{bmatrix}$.

5. Figure 36.7 shows a weighted graph G_w , where the parameters a, b, c are the weights of the edges, and where the vertices are labeled $1, \dots, 4$, as indicated in that figure. The corresponding Laplacian matrix is $L(G_w) =$

$$\begin{bmatrix} b & 0 & 0 & -b \\ 0 & a+c & -c & -a \\ 0 & -c & a+c & -a \\ -b & -a & -a & 2a+b \end{bmatrix}. \text{ The eigenvalues of } L(G_w) \text{ are } 0,$$

$a+2c$, and $\frac{3a+2b \pm \sqrt{9a^2-4ab+4b^2}}{2}$, so that $\alpha(G_w) = \min\{a+2c, \frac{3a+2b-\sqrt{9a^2-4ab+4b^2}}{2}\}$. If $\alpha(G_w) = a+2c$,

$$\text{then } \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} \text{ is a Fiedler vector for } G_w, \text{ while if } \alpha(G_w) = \frac{3a+2b-\sqrt{9a^2-4ab+4b^2}}{2}, \text{ then } \begin{bmatrix} \frac{3a+\sqrt{9a^2-4ab+4b^2}}{2} \\ -\frac{a+2b+\sqrt{9a^2-4ab+4b^2}}{4} \\ -\frac{a+2b+\sqrt{9a^2-4ab+4b^2}}{4} \\ b-a \end{bmatrix}$$

is a Fiedler vector for G_w .

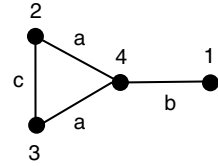


FIGURE 36.7 The weighted graph G_w in Example 5.

36.5 Absolute Algebraic Connectivity for Simple Graphs

The term *graph* means *simple graph* in this section.

Definitions:

Let $G = (V, E)$ be a graph on $n \geq 2$ vertices. Let W denote the set of all nonnegative weightings of G with the property that for each weighting w , we have $\sum_{e \in E} w_e = |E|$ (observe that here we relax the restriction that each edge weight be positive, and allow zero weights). The **absolute algebraic connectivity** of G is denoted by $\hat{\alpha}(G)$, and is given by $\hat{\alpha}(G) = \max\{\alpha(G_w) | w \in W\}$.

Let $T = (V, E)$ be a tree. For each vertex u of T , define $S(u) = \sum_{k \in V} (d(u, k))^2$.

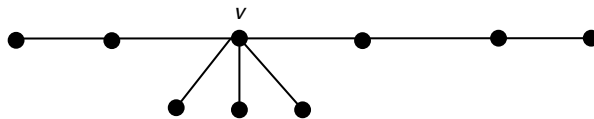
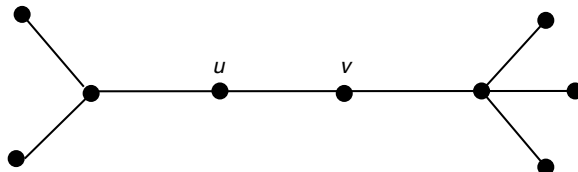
Facts:

1. [Fie90] For a graph G , $\hat{\alpha}(G) = 0$ if and only if G is disconnected.
2. [Fie90] If G is a graph and Ω is its automorphism group, then $\hat{\alpha}(G) = \max\{\alpha(G_w) | w \in W_0\}$, where W_0 is the subclass of W consisting of weightings for which equivalent edges in Ω have the same weight.
3. [KP02] If G is a graph with m edges, and H is the graph formed from G by adding an edge not already present in G , then $\frac{m+1}{m} \hat{\alpha}(G) \leq \hat{\alpha}(H)$.
4. [Fie90] Let T be a tree on n vertices. Then one of the following two cases occurs:

- There is a vertex v of T such that for any vertex $k \neq v$, $S(v) \leq S(k) - n$. In this case, $\hat{\alpha}(T) = \frac{n-1}{S(v)}$.
- There is an edge $\{u, v\}$ of T such that $|S(u) - S(v)| < n$. In this case, $\hat{\alpha}(T) = \frac{4n(n-1)}{4nS(u) - (n+S(u)-S(v))^2}$.

Weightings of trees that yield the maximum value $\hat{\alpha}$ are also discussed in [Fie90].

5. [Fie93] If T is a tree on n vertices, then $\frac{12}{n(n+1)} \leq \hat{\alpha}(T) \leq 1$. Equality holds in the lower bound if $T = P_n$, and equality holds in the upper bound if $T = K_{1,n-1}$.
6. [KP02] Suppose that G is a graph on $n \geq 7$ vertices that has a cut-vertex. Then $\hat{\alpha}(G) \leq \left(\frac{n^2-3n+4}{n-3} \right) \left(1 - \frac{4}{2(n-1)-(n-3)\sqrt{2(n-2)/(n-1)}} \right)$. Further, the upper bound is attained by the graph formed by appending a pendent vertex at a vertex of K_{n-1} ; a weighting of that graph that yields the maximum value $\hat{\alpha}$ is also given in [KP02].

FIGURE 36.8 The tree $P(9, 5)$.FIGURE 36.9 The tree $T(2, 3, 4)$.**Examples:**

1. The absolute algebraic connectivity of K_n is n .
2. If $m \leq n$ and $2 \leq n$, then $\hat{\alpha}(K_{m,n}) = m$.
3. [KP02] Let e be an edge of K_n . Then $\hat{\alpha}(K_n - e) = \frac{(n-2)(n^2-n-2)}{n^2-3n+4}$.
4. Consider the tree $P(9, 5)$ shown in Figure 36.8; we have $S(v) = 22$, and $S(v) \leq S(k) - 9$ for any vertex $k \neq v$, so that the first case of Fact 4 above holds. Hence, $\hat{\alpha}(P(9, 5)) = \frac{4}{11}$.
5. Consider the tree $T(2, 3, 4)$ pictured in Figure 36.9. We have $S(u) = 41$ and $S(v) = 36$, so that the second case of Fact 4 above holds. Hence, $\hat{\alpha}(T(2, 3, 4)) = \frac{9}{40}$.

36.6 Generalized Laplacians and Multiplicity

The term *graph* means *simple graph* in this section.

Definitions:

Let $G = (V, E)$ be a graph on $n \geq 2$ vertices. A **generalized Laplacian** of G is a symmetric $n \times n$ matrix M with the property that for each $i \neq j$, $m_{i,j} < 0$ if $\{i, j\} \in E$ and $m_{i,j} = 0$ if $\{i, j\} \notin E$.

We only consider generalized Laplacian matrices for graphs on at least two vertices.

For any symmetric matrix B , let $\tau(B)$ denote its smallest eigenvalue, let $\mu(B)$ denote its second smallest eigenvalue, and let $\lambda(B)$ denote its largest eigenvalue.

Let G be a graph, and let v be a vertex of G . Suppose that M is a generalized Laplacian matrix for G . Let $C_1, \dots, C_k$ denote the components of $G - v$; a component C_j is called a **Perron component at v** if $\tau(M[C_j])$ is minimal among all $\tau(M[C_i])$, $i = 1, \dots, k$.

Facts:

1. [BKP01] Let G be a connected graph and let v be a cut-vertex of G . Suppose that M is a generalized Laplacian matrix for G . If there are $p \geq 2$ Perron components at v , say $C_1, \dots, C_p$, then $\mu(M) = \tau(M[C_1])$. Further, $\mu(M)$ is an eigenvalue of M of multiplicity $p - 1$.
2. [BKP01] Let G be a connected graph, and let v be a cut-vertex of G . Let M be a generalized Laplacian matrix for G . Suppose that there is a unique Perron component A_0 at v ; denote the other connected components at v by $B_1, \dots, B_k$, and set $A_1 = G \setminus A_0$. Let $\mathbf{z}$ be an eigenvector of M corresponding to $\tau(M)$, and let $\mathbf{z}_0, \mathbf{z}_1$ be the subvectors of $\mathbf{z}$ corresponding to the vertices in A_0, A_1 , respectively. For each $i = 1, \dots, k$, denote the principal submatrix of M corresponding to

the vertices of B_i by $M[B_i]$. Let $S = (M[B_1])^{-1} \oplus \dots \oplus (M[B_k])^{-1} \oplus [0]$, and let M_0 denote the principal submatrix of M corresponding to the vertices of A_0 . There is a unique $\gamma > 0$ such that $\lambda(M_0^{-1} - \gamma \mathbf{z}_0 \mathbf{z}_0^T) = \lambda(S + \gamma \mathbf{z}_1 \mathbf{z}_1^T) = 1/\mu(M)$. Further, the multiplicity of $\mu(M)$ as an eigenvalue of M coincides with the multiplicity of $1/\mu(M)$ as an eigenvalue of $M_0^{-1} - \gamma \mathbf{z}_0 \mathbf{z}_0^T$.

Examples:

1. Consider the generalized Laplacian matrix $M = \begin{bmatrix} 0 & -1 & -1 & -1 & -2 \\ -1 & 3 & -1 & 0 & 0 \\ -1 & -1 & 3 & 0 & 0 \\ -1 & 0 & 0 & 2 & 0 \\ -2 & 0 & 0 & 0 & 2 \end{bmatrix}$. There are three

Perron components at vertex 1, induced by the vertex sets $\{2, 3\}$, $\{4\}$, and $\{5\}$. We have $\mu(M) = 2$, which is an eigenvalue of multiplicity two.

2. Consider the generalized Laplacian matrix $M = \begin{bmatrix} -3 & -1 & -1 & -1 & -2 \\ -1 & 2 & -1 & 0 & 0 \\ -1 & -1 & 2 & 0 & 0 \\ -1 & 0 & 0 & 2 & 0 \\ -2 & 0 & 0 & 0 & 2 \end{bmatrix}$. The unique Perron

component at vertex 1 is induced by the vertex set $\{2, 3\}$. We have $\mu(M) = \frac{-3+\sqrt{29}}{2}$, which is a simple eigenvalue.

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III

Numerical Methods

Numerical Methods for Linear Systems

- 37 Vector and Matrix Norms, Error Analysis, Efficiency,
and Stability *Ralph Byers and Biswa Nath Datta* 37-1
- 38 Matrix Factorizations and Direct Solution of Linear
Systems *Christopher Beattie* 38-1
- 39 Least Squares Solution of Linear Systems *Per Christian Hansen
and Hans Bruun Nielsen* 39-1
- 40 Sparse Matrix Methods *Esmond G. Ng* 40-1
- 41 Iterative Solution Methods for Linear Systems *Anne Greenbaum* 41-1

Numerical Methods for Eigenvalues

- 42 Symmetric Matrix Eigenvalue Techniques *Ivan Slapničar* 42-1
- 43 Unsymmetric Matrix Eigenvalue Techniques *David S. Watkins* 43-1
- 44 The Implicitly Restarted Arnoldi Method *D. C. Sorensen* 44-1
- 45 Computation of the Singular Value Decomposition *Alan Kaylor Cline
and Inderjit S. Dhillon* 45-1
- 46 Computing Eigenvalues and Singular Values to High Relative Accuracy
Zlatko Drmač 46-1

Computational Linear Algebra

- 47 Fast Matrix Multiplication *Dario A. Bini* 47-1
- 48 Structured Matrix Computations *Michael Ng* 48-1
- 49 Large-Scale Matrix Computations *Roland W. Freund* 49-1

Numerical Methods for Linear Systems

37	Vector and Matrix Norms, Error Analysis, Efficiency, and Stability	<i>Ralph Byers and Biswa Nath Datta</i>	37-1
	Vector Norms • Vector Seminorms • Matrix Norms • Conditioning and Condition Numbers • Conditioning of Linear Systems • Floating Point Numbers • Algorithms and Efficiency • Numerical Stability and Instability		
38	Matrix Factorizations and Direct Solution of Linear Systems	<i>Christopher Beattie</i>	38-1
	Perturbations of Linear Systems • Triangular Linear Systems • Gauss Elimination and LU Decomposition • Orthogonalization and QR Decomposition • Symmetric Factorizations		
39	Least Squares Solution of Linear Systems	<i>Per Christian Hansen and Hans Bruun Nielsen</i>	39-1
	Basic Concepts • Least Squares Data Fitting • Geometric and Algebraic Aspects • Orthogonal Factorizations • Least Squares Algorithms • Sensitivity • Up- and Dondating of QR Factorization • Damped Least Squares • Rank Revealing Decompositions		
40	Sparse Matrix Methods	<i>Esmond G. Ng</i>	40-1
	Introduction • Sparse Matrices • Sparse Matrix Factorizations • Modeling and Analyzing Fill • Effect of Reorderings		
41	Iterative Solution Methods for Linear Systems	<i>Anne Greenbaum</i>	41-1
	Krylov Subspaces and Preconditioners • Optimal Krylov Space Methods for Hermitian Problems • Optimal and Nonoptimal Krylov Space Methods for Non-Hermitian Problems • Preconditioners • Preconditioned Algorithms • Convergence Rates of CG and MINRES • Convergence Rate of GMRES • Inexact Preconditioners and Finite Precision Arithmetic, Error Estimation and Stopping Criteria, Text and Reference Books		

37

Vector and Matrix Norms, Error Analysis, Efficiency, and Stability

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37.1	Vector Norms	37-2
37.2	Vector Seminorms	37-3
37.3	Matrix Norms	37-4
37.4	Conditioning and Condition Numbers.....	37-7
37.5	Conditioning of Linear Systems.....	37-9
37.6	Floating Point Numbers	37-11
37.7	Algorithms and Efficiency	37-16
37.8	Numerical Stability and Instability	37-18
	References	37-22

Calculations are subject to errors. There may be modeling errors, measurement errors, manufacturing errors, noise, equipment is subject to wear and damage, etc. In preparation for computation, data must often be perturbed by rounding it to fit a particular finite precision, floating-point format. Further errors may be introduced during a computation by using finite precision arithmetic and by truncating an infinite process down to a finite number of steps.

This chapter outlines aspects of how such errors affect the results of a mathematical computation with an emphasis on matrix computations. Topics include:

- Vector and matrix norms and seminorms that are often used to analyze errors and express error bounds.
- Floating point arithmetic.
- Condition numbers that measure how much data errors may affect computational results.
- Numerical stability, an assessment of whether excessive amounts of data may be lost to rounding errors during a finite precision computation.

For elaborative discussions of vector and matrix norms, consult [HJ85]. See [Ove01] for a textbook introduction to IEEE standard floating point arithmetic. Complete details of the standard are available in [IEEE754] and [IEEE854]. Basic concepts of numerical stability and conditioning appear in numerical linear algebra books, e.g., [GV96], [Ste98], [TB97], and [Dat95]. The two particularly authoritative books on these topics are the classical book by Wilkinson [Wil65] and the more recent one by Higham [Hig96]. For perturbation analysis, see the classic monograph [Kat66] and the more modern treatments in [SS90] and [Bha96].

The set $\mathbb{C}^n = \mathbb{C}^{n \times 1}$ is the complex vector space of n -row, 1-column matrices, and $\mathbb{R}^n = \mathbb{R}^{n \times 1}$ is the real vector space of n -row, 1-column matrices. Unless otherwise specified, $\mathbb{F}^n$ is either $\mathbb{R}^n$ or $\mathbb{C}^n$, $\mathbf{x}$ and $\mathbf{y}$ are members of $\mathbb{F}^n$, and $\alpha \in \mathbb{F}$ is a scalar $\alpha \in \mathbb{R}$ or $\alpha \in \mathbb{C}$, respectively. For $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}^T$ is the one row n -column transpose of $\mathbf{x}$. For $\mathbf{x} \in \mathbb{C}^n$, $\mathbf{x}^*$ is the one row n -column complex-conjugate-transpose of $\mathbf{x}$. A and B are members of $\mathbb{F}^{m \times n}$. For $A \in \mathbb{R}^{m \times n}$, $A^* \in \mathbb{R}^{n \times m}$ is the transpose of A . For $A \in \mathbb{C}^{m \times n}$, $A^* \in \mathbb{C}^{n \times m}$ is the complex-conjugate-transpose of A .

37.1 Vector Norms

Most uses of vector norms involve $\mathbb{R}^n$ or $\mathbb{C}^n$, so the focus of this section is on those vector spaces. However, the definitions given here can be extended in the obvious way to any finite dimensional real or complex vector space.

Let $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ and $\alpha \in \mathbb{F}$, where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$.

Definitions:

A vector **norm** is a real-valued function on $\mathbb{F}^n$ denoted $\|\mathbf{x}\|$ with the following properties for all $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ and all scalars $\alpha \in \mathbb{F}$.

- **Positive definiteness:** $\|\mathbf{x}\| \geq 0$ and $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x}$ is the zero vector.
- **Homogeneity:** $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$.
- **Triangle inequality:** $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

For $\mathbf{x} = [x_1, x_2, x_3, \dots, x_n]^* \in \mathbb{F}^n$, the following are commonly encountered vector norms.

- **Sum-norm or 1-norm:** $\|\mathbf{x}\|_1 = |x_1| + |x_2| + \dots + |x_n|$.
- **Euclidean norm or 2-norm:** $\|\mathbf{x}\|_2 = \sqrt{|x_1|^2 + |x_2|^2 + \dots + |x_n|^2}$.
- **Sup-norm or ∞ -norm:** $\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|$.
- **Hölder norm or p -norm:** For $p \geq 1$, $\|\mathbf{x}\|_p = (|x_1|^p + \dots + |x_n|^p)^{\frac{1}{p}}$.

If $\|\cdot\|$ is a vector norm on $\mathbb{F}^n$ and $M \in \mathbb{F}^{n \times n}$ is a nonsingular matrix, then $\|\mathbf{y}\|_M \equiv \|M\mathbf{y}\|$ is an **M -norm** or **energy norm**. (Note that this notation is ambiguous, since $\|\cdot\|$ is not specified; it either doesn't matter or must be stated explicitly when used.)

A vector norm $\|\cdot\|$ is **absolute** if for all $\mathbf{x} \in \mathbb{F}^n$, $\|\mathbf{x}\| = \|\mathbf{x}\|$, where $|\mathbf{x}| = [|x_1|, \dots, |x_n]|^*$.

A vector norm $\|\cdot\|$ is **monotone** if for all $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, $|\mathbf{x}| \leq |\mathbf{y}|$ implies $\|\mathbf{x}\| \leq \|\mathbf{y}\|$.

A vector norm $\|\cdot\|$ is **permutation invariant** if $\|P\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x} \in \mathbb{R}^n$ and all permutation matrices $P \in \mathbb{R}^{n \times n}$.

Let $\|\cdot\|$ be a vector norm. The **dual norm** is defined by $\|\mathbf{y}\|^D = \max_{\mathbf{x} \neq 0} \frac{|\mathbf{y}^* \mathbf{x}|}{\|\mathbf{x}\|}$.

The unit disk corresponding to a vector norm $\|\cdot\|$ is the set $\{\mathbf{x} \in \mathbb{F}^n \mid \|\mathbf{x}\| \leq 1\}$.

The unit sphere corresponding to a vector norm $\|\cdot\|$ is the set $\{\mathbf{x} \in \mathbb{F}^n \mid \|\mathbf{x}\| = 1\}$.

Facts:

For proofs and additional background, see, for example, [HJ85, Chap. 5].

Let $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ and $\alpha \in \mathbb{F}$, where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$.

1. The commonly encountered norms, $\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_\infty, \|\cdot\|_p$, are permutation invariant, absolute, monotone vector norms.
2. If $M \in \mathbb{F}^{n \times n}$ is a nonsingular matrix and $\|\cdot\|$ is a vector norm, then the M -norm $\|\cdot\|_M$ is a vector norm.
3. If $\|\cdot\|$ is a vector norm, then $|\|\mathbf{x}\| - \|\mathbf{y}\|| \leq \|\mathbf{x} - \mathbf{y}\|$.
4. A sum of vector norms is a vector norm.
5. $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{x}_*$ if and only if in any norm $\lim_{k \rightarrow \infty} \|\mathbf{x}_k - \mathbf{x}_*\| = 0$.

6. Cauchy–Schwartz inequality:

(a) $|\mathbf{x}^* \mathbf{y}| \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$.

(b) $|\mathbf{x}^* \mathbf{y}| = \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$ if and only if there exist scalars α and β , not both zero, for which $\alpha \mathbf{x} = \beta \mathbf{y}$.

7. **Hölder inequality:** If $p \geq 1$ and $q \geq 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$, then $|\mathbf{x}^* \mathbf{y}| \leq \|\mathbf{x}\|_p \|\mathbf{y}\|_q$.8. If $\|\cdot\|$ is a vector norm on $\mathbb{F}^n$, then its dual $\|\cdot\|^D$ is also a vector norm on $\mathbb{F}^n$, and $\|\cdot\|^D = \|\cdot\|$.9. If $p > 0$ and $q > 0$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$, then $\|\cdot\|_p^D = \|\cdot\|_q$. In particular, $\|\cdot\|_2^D = \|\cdot\|_2$. Also, $\|\cdot\|_1^D = \|\cdot\|_\infty$.10. If $\|\cdot\|$ is a vector norm on $\mathbb{F}^n$, then for any $\mathbf{x} \in \mathbb{F}^n$, $|\mathbf{x}^* \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|^D$.

11. A vector norm is absolute if and only if it is monotone.

12. **Equivalence of norms:** All vector norms on $\mathbb{F}^n$ are equivalent in the sense that for any two vector norms $\|\cdot\|_\mu$ and $\|\cdot\|_\nu$ there constants $\alpha > 0$ and $\beta > 0$ such that for all $\mathbf{x} \in \mathbb{F}^n$, $\alpha \|\mathbf{x}\|_\mu \leq \|\mathbf{x}\|_\nu \leq \beta \|\mathbf{x}\|_\mu$. The constants α and β are independent of $\mathbf{x}$ but typically depend on the dimension n .

In particular,

(a) $\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1 \leq \sqrt{n} \|\mathbf{x}\|_2$.

(b) $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \sqrt{n} \|\mathbf{x}\|_\infty$.

(c) $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_1 \leq n \|\mathbf{x}\|_\infty$.

13. A set $D \subset \mathbb{F}^n$ is the unit disk of a vector norm if and only if it has the following properties.(a) **Point-wise bounded:** For every vector $\mathbf{x} \in \mathbb{F}^n$ there is a number $\delta > 0$ for which $\delta \mathbf{x} \notin D$.(b) **Absorbing:** For every vector $\mathbf{x} \in \mathbb{F}^n$ there is a number $\tau > 0$ for which $|\alpha| \leq \tau$ implies $\alpha \mathbf{x} \in D$.(c) **Convex:** For every pair of vectors $\mathbf{x}, \mathbf{y} \in D$ and every number t , $0 \leq t \leq 1$, $t\mathbf{x} + (1-t)\mathbf{y} \in D$.**Examples:**1. Let $\mathbf{x} = [1, 1, -2]^*$. Then $\|\mathbf{x}\|_1 = 4$, $\|\mathbf{x}\|_2 = \sqrt{1^2 + 1^2 + (-2)^2} = \sqrt{6}$, and $\|\mathbf{x}\|_\infty = 2$.2. Let $M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Using the 1-norm, $\left\| \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\|_M = \left\| \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\|_1 = 6$.

37.2 Vector Seminorms

Definitions:A **vector seminorm** is a real-valued function on $\mathbb{F}^n$, denoted $v(\mathbf{x})$, with the following properties for all $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ and all scalars $\alpha \in \mathbb{F}$.1. **Positiveness:** $v(\mathbf{x}) \geq 0$.2. **Homogeneity:** $\|\alpha \mathbf{x}\| = |\alpha| \|\mathbf{x}\|$.3. **Triangle inequality:** $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.Vector norms are *a fortiori* also vector seminorms.The unit disk corresponding to a vector seminorm $\|\cdot\|$ is the set $\{\mathbf{x} \in \mathbb{F}^n \mid v(\mathbf{x}) \leq 1\}$.The unit sphere corresponding to a vector seminorm $\|\cdot\|$ is the set $\{\mathbf{x} \in \mathbb{F}^n \mid v(\mathbf{x}) = 1\}$.**Facts:**

For proofs and additional background, see, for example, [HJ85, Chap. 5].

Let $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$ and $\alpha \in \mathbb{F}$, where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$.1. $v(0) = 0$.2. $v(\mathbf{x} - \mathbf{y}) \geq |v(\mathbf{x}) - v(\mathbf{y})|$.

3. A sum of vector seminorms is a vector seminorm. If one of the summands is a vector norm, then the sum is a vector norm.
4. A set $D \subset \mathbb{F}^n$ is the unit disk of a seminorm if and only if it has the following properties.
 - (a) **Absorbing:** For every vector $\mathbf{x} \in \mathbb{F}^n$ there is a number $\tau > 0$ for which $|\alpha| \leq \tau$ implies $\alpha\mathbf{x} \in D$.
 - (b) **Convex:** For every pair of vectors $\mathbf{x}, \mathbf{y} \in D$ and every number $t, 0 \leq t \leq 1$, $t\mathbf{x} + (1-t)\mathbf{y} \in D$.

Examples:

1. For $\mathbf{x} = [x_1, x_2, x_3, \dots, x_n]^T \in \mathbb{F}^n$, the function $\nu(\mathbf{x}) = |x_1|$ is a vector seminorm that is not a vector norm. For $n \geq 2$, this seminorm is not equivalent to any vector norm $\|\cdot\|$, since $\|\mathbf{e}_2\| > 0$ but $\nu(\mathbf{e}_2) = 0$, for $\mathbf{e}_2 = [0, 1, 0, \dots, 0]^T$.

37.3 Matrix Norms

Definitions:

A **matrix norm** is a family of real-valued functions on $\mathbb{F}^{m \times n}$ for all positive integers m and n , denoted uniformly by $\|A\|$ with the following properties for all matrices A and B and all scalars $\alpha \in \mathbb{F}$.

- **Positive definiteness:** $\|A\| \geq 0$; $\|A\| = 0$ only if $A = 0$.
- **Homogeneity:** $\|\alpha A\| = |\alpha| \|A\|$.
- **Triangle inequality:** $\|A + B\| \leq \|A\| + \|B\|$, where A and B are compatible for matrix addition.
- **Consistency:** $\|AB\| \leq \|A\| \|B\|$, where A and B are compatible for matrix multiplication.

If $\|\cdot\|$ is a family of vector norms on $\mathbb{F}^n$ for $n = 1, 2, 3, \dots$, then the matrix norm on $\mathbb{F}^{m \times n}$ **induced** by (or **subordinate** to) $\|\cdot\|$ is $\|A\| = \max_{\mathbf{x} \neq 0} \frac{\|A\mathbf{x}\|}{\|\mathbf{x}\|}$. Induced matrix norms are also called **operator norms** or **natural norms**. The matrix norm $\|A\|_p$ denotes the norm induced by the Hölder vector norm $\|\mathbf{x}\|_p$. The following are commonly encountered matrix norms.

- **Maximum absolute column sum norm:** $\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|$.
- **Spectral norm:** $\|A\|_2 = \sqrt{\rho(A^*A)}$, where $\rho(A^*A)$ is the largest eigenvalue of A^*A .
- **Maximum absolute row sum norm:** $\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$.
- **Euclidean norm or Frobenius norm:** $\|A\|_F = \sqrt{\sum_{i,j=1}^n |a_{ij}|^2}$.

Let $\mathcal{M} = \{M_n \in F^{n \times n} : n \geq 1\}$ be a family of nonsingular matrices and let $\|\cdot\|$ be a family of vector norms. Define a family of vector norms by $\|\mathbf{x}\|_{\mathcal{M}}$ for $\mathbf{x} \in \mathbb{F}^n$ by $\|\mathbf{x}\|_{\mathcal{M}} = \|M_n \mathbf{x}\|$. This family of vector norms is also called the **$\mathcal{M}$ -norm** and denoted by $\|\cdot\|_{\mathcal{M}}$. (Note that this notation is ambiguous, since $\|\cdot\|$ is not specified; it either does not matter or must be stated explicitly when used.)

A matrix norm $\|\cdot\|$ is **minimal** if for any matrix norm $\|\cdot\|_v$, $\|A\|_v \leq \|A\|$ for all $A \in F^{n \times n}$ implies $\|\cdot\|_v = \|\cdot\|$.

A matrix norm is **absolute** if as a vector norm, each member of the family is absolute.

Facts:

For proofs and additional background, see, for example, [HJ85, Chap. 5]. Let $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, $A, B \in \mathbb{F}^{m \times n}$, and $\alpha \in \mathbb{F}$, where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$.

1. A matrix norm is a family of vector norms, but not every family of vector norms is a matrix norm (see Example 2).
2. The commonly encountered norms, $\|\cdot\|_1$, $\|\cdot\|_2$, $\|\cdot\|_\infty$, $\|\cdot\|_F$, and norms induced by vector norms are matrix norms. Furthermore,
 - (a) $\|A\|_1$ is the matrix norm induced by the vector norm $\|\cdot\|_1$.
 - (b) $\|A\|_2$ is the matrix norm induced by the vector norm $\|\cdot\|_2$.
 - (c) $\|A\|_\infty$ is the matrix norm induced by the vector norm $\|\cdot\|_\infty$.
 - (d) $\|A\|_F$ is not induced by any vector norm.
 - (e) If $\mathcal{M} = \{M_n\}$ is a family of nonsingular matrices and $\|\cdot\|$ is an induced matrix norm, then for $A \in \mathbb{F}^{m \times n}$, $\|A\|_{\mathcal{M}} = \|M_m A M_n^{-1}\|$.
3. If $\|\cdot\|$ is the matrix norm induced by a family of vector norms $\|\cdot\|$, then $\|I_n\| = 1$ for all positive integers n (where I_n is the $n \times n$ identity matrix).
4. If $\|\cdot\|$ is the matrix norm induced by a family of vector norms $\|\cdot\|$, then for all $A \in \mathbb{F}^{m \times n}$ and all $\mathbf{x} \in \mathbb{F}^n$, $\|A\mathbf{x}\| \leq \|A\| \|\mathbf{x}\|$.
5. For all $A \in \mathbb{F}^{m \times n}$ and all $\mathbf{x} \in \mathbb{F}^n$, $\|A\mathbf{x}\|_F \leq \|A\|_F \|\mathbf{x}\|_2$.
6. $\|\cdot\|_1$, $\|\cdot\|_\infty$, $\|\cdot\|_F$ are absolute norms. However, for some matrices A , $\| |A| \|_2 \neq \|A\|_2$ (see Example 3).
7. A matrix norm is minimal if and only if it is an induced norm.
8. All matrix norms are equivalent in the sense that for any two matrix norms $\|\cdot\|_\mu$ and $\|\cdot\|_\nu$, there exist constants $\alpha > 0$ and $\beta > 0$ such that for all $A \in \mathbb{F}^{m \times n}$, $\alpha \|A\|_\mu \leq \|A\|_\nu \leq \beta \|A\|_\mu$. The constants α and β are independent of A but typically depend on n and m . In particular,
 - (a) $\frac{1}{\sqrt{n}} \|A\|_\infty \leq \|A\|_2 \leq \sqrt{m} \|A\|_\infty$.
 - (b) $\|A\|_2 \leq \|A\|_F \leq \sqrt{n} \|A\|_2$.
 - (c) $\frac{1}{\sqrt{m}} \|A\|_1 \leq \|A\|_2 \leq \sqrt{n} \|A\|_1$.
9. $\|A\|_2 \leq \sqrt{\|A\|_1 \|A\|_\infty}$.
10. $\|AB\|_F \leq \|A\|_F \|B\|_2$ and $\|AB\|_F \leq \|A\|_2 \|B\|_F$ whenever A and B are compatible for matrix multiplication.
11. $\|A\|_2 \leq \|A\|_F$ and $\|A\|_2 = \|A\|_F$ if and only if A has rank less than or equal to 1.
12. If $A = \mathbf{xy}^*$ for some $\mathbf{x} \in \mathbb{F}^n$ and $\mathbf{y} \in \mathbb{F}^m$, then $\|A\|_2 = \|A\|_F = \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$.
13. $\|A\|_2 = \|A^*\|_2$ and $\|A\|_F = \|A^*\|_F$.
14. If $U \in \mathbb{F}^{n \times n}$ is a unitary matrix, i.e., if $U^* = U^{-1}$, then the following hold.
 - (a) $\|U\|_2 = 1$ and $\|U\|_F = \sqrt{n}$.
 - (b) If $A \in \mathbb{F}^{m \times n}$, then $\|AU\|_2 = \|A\|_2$ and $\|AU\|_F = \|A\|_F$.
 - (c) If $A \in \mathbb{F}^{n \times m}$, then $\|UA\|_2 = \|A\|_2$ and $\|UA\|_F = \|A\|_F$.
15. For any matrix norm $\|\cdot\|$ and any $A \in \mathbb{F}^{n \times n}$, $\rho(A) \leq \|A\|$, where $\rho(A)$ is the spectral radius of A . This need not be true for a vector norm on matrices (see Example 2).
16. For any $A \in \mathbb{F}^{n \times n}$ and $\varepsilon > 0$, there exists a matrix norm $\|\cdot\|$ such that $\|A\| < \rho(A) + \varepsilon$. A method for finding such a norm is given in Example 5.
17. For any matrix norm $\|\cdot\|$ and $A \in \mathbb{F}^{n \times n}$, $\lim_{k \rightarrow \infty} \|A^k\|^{1/k} = \rho(A)$.
18. For $A \in \mathbb{F}^{n \times n}$, $\lim_{k \rightarrow \infty} A^k = 0$ if and only if $\rho(A) < 1$.

Examples:

1. If $A = \begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix}$, then $\|A\|_1 = 6$, $\|A\|_\infty = 7$, $\|A\|_2 = \sqrt{15 + \sqrt{221}}$, and $\|A\|_F = \sqrt{30}$.
2. The family of matrix functions defined for $A \in \mathbb{F}^{m \times n}$ by

$$v(A) = \max_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}} |a_{ij}|$$

is *not* a matrix norm because consistency fails. For example, if $J = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, then $v(J^2) = 2 > 1 = v(J)v(J)$. Note that v is a family of vector norms on matrices (it is the ∞ norm on the n^2 -tuple of entries), and $v(J) = 1 < 2 = \rho(J)$.

3. If $A = \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix}$, then $\|A\|_2 = 5$ but $\| |A| \|_2 = 7$.
4. If A is perturbed by an error matrix E and U is unitary (i.e., $U^* = U^{-1}$), then $U(A+E) = UA+UE$ and $\|UE\|_2 = \|E\|_2$. Numerical analysts often use unitary matrices in numerical algorithms because multiplication by unitary matrices does not magnify errors.
5. Given $A \in F^{n \times n}$ and $\varepsilon > 0$, we show how an $\mathcal{M}$ -norm can be constructed such that $\|A\|_{\mathcal{M}} < \rho + \varepsilon$, where ρ is the spectral radius of A . The procedure below determines M_n where $A \in F^{n \times n}$. The

procedure is illustrated with the matrix $A = \begin{bmatrix} -38 & 13 & 52 \\ 3 & 0 & -4 \\ -30 & 10 & 41 \end{bmatrix}$ and with $\varepsilon = 0.1$. The norm used

to construct the $\mathcal{M}$ -norm will be the 1-norm; note the 1-norm of $A = 97$.

- (a) Determine ρ : The characteristic polynomial of A is $p_A(x) = \det(A - xI) = x^3 - 3x^2 + 3x - 1 = (x - 1)^3$, so $\rho = 1$.
- (b) Find a unitary matrix U such that $T = UAU^*$ is triangular. Using the method in Example 5 of Chapter 7.1, we find

$$U = \begin{bmatrix} \frac{1}{\sqrt{10}} & \frac{6}{\sqrt{65}} & -\frac{3}{\sqrt{26}} \\ \frac{3}{\sqrt{10}} & -\frac{2}{\sqrt{65}} & \frac{1}{\sqrt{26}} \\ 0 & \sqrt{\frac{5}{13}} & 2\sqrt{\frac{2}{13}} \end{bmatrix} \approx \begin{bmatrix} 0.316228 & 0.744208 & -0.588348 \\ 0.948683 & -0.248069 & 0.196116 \\ 0 & 0.620174 & 0.784465 \end{bmatrix} \text{ and}$$

$$T = U^*AU = \begin{bmatrix} 1 & 0 & 2\sqrt{65} \\ 0 & 1 & 26\sqrt{10} \\ 0 & 0 & 1 \end{bmatrix} \approx \begin{bmatrix} 1 & 0 & 16.1245 \\ 0 & 1 & 82.2192 \\ 0 & 0 & 1 \end{bmatrix}.$$

- (c) Find a diagonal matrix $\text{diag}(1, \alpha, \alpha^2, \dots, \alpha^{n-1})$ such that $\|DTD^{-1}\|_1 < \rho + \varepsilon$ (this is always possible, since $\lim_{\alpha \rightarrow \infty} \|DTD^{-1}\|_1 = \rho$).

$$\text{In the example, for } \alpha = 1000, DTD^{-1} \approx \begin{bmatrix} 1 & 0 & 0.0000161245 \\ 0 & 1 & 0.0822192 \\ 0 & 0 & 1 \end{bmatrix} \text{ and } \|DTD^{-1}\|_1 \approx$$

$$1.08224 < 1.1.$$

- (d) Then $\|DU^*AUD^{-1}\|_1 < \rho + \varepsilon$. That is, $\|A\|_{\mathcal{M}} < 2.1$, where $M_3 = DU^*$

$$\approx \begin{bmatrix} 0.316228 & 0.948683 & 0 \\ 744.208 & -248.069 & 620.174 \\ -588348 & 196116 & 784465 \end{bmatrix}.$$

37.4 Conditioning and Condition Numbers

Data have limited precision. Measurements are inexact, equipment wears, manufactured components meet specifications only to some error tolerance, floating point arithmetic introduces errors. Consequently, the results of nontrivial calculations using data of limited precision also have limited precision. This section summarizes the topic of **conditioning**: How much errors in data can affect the results of a calculation.

(See [Ric66] for an authoritative treatment of conditioning.)

Definitions:

Consider a computational problem to be the task of evaluating a function $P : \mathbb{R}^n \rightarrow \mathbb{R}^m$ at a nominal data point $\mathbf{z} \in \mathbb{R}^n$, which, because data errors are ubiquitous, is known only to within a small relative-to- $\|\mathbf{z}\|$ error ε .

If $\hat{\mathbf{z}} \in \mathbb{R}^n$ is an approximation to $\mathbf{z} \in \mathbb{R}^n$, the **absolute error** in $\hat{\mathbf{z}}$ is $\|\mathbf{z} - \hat{\mathbf{z}}\|$ and the **relative error** in $\hat{\mathbf{z}}$ is $\|\mathbf{z} - \hat{\mathbf{z}}\|/\|\mathbf{z}\|$. If $\mathbf{z} = 0$, then the relative error is undefined.

The data $\mathbf{z}$ are **well-conditioned** if small relative perturbations of $\mathbf{z}$ cause small relative perturbations of $P(\mathbf{z})$. The data are **ill-conditioned** or **badly conditioned** if some small relative perturbation of $\mathbf{z}$ causes a large relative perturbation of $P(\mathbf{z})$. Precise meanings of “small” and “large” are dependent on the precision required in the context of the computational task.

Note that it is the data $\mathbf{z}$ — not the solution $P(\mathbf{z})$ — that is ill-conditioned or well-conditioned.

If $\mathbf{z} \neq 0$ and $P(\mathbf{z}) \neq 0$, then the **relative condition number**, or simply **condition number** $\mathbf{cond}(\mathbf{z}) = \mathbf{cond}_P(\mathbf{z})$ of the data $\mathbf{z} \in \mathbb{R}^n$ with respect to the computational task of evaluating $P(\mathbf{z})$ may be defined as

$$\mathbf{cond}_P(\mathbf{z}) = \limsup_{\varepsilon \rightarrow 0} \left\{ \left(\frac{\|P(\mathbf{z} + \delta\mathbf{z}) - P(\mathbf{z})\|}{\|P(\mathbf{z})\|} \right) \left(\frac{\|\mathbf{z}\|}{\|\delta\mathbf{z}\|} \right) \mid \|\delta\mathbf{z}\| \leq \varepsilon \right\}. \quad (37.1)$$

Sometimes it is useful to extend the definition to $\mathbf{z} = 0$ or to an isolated root of $P(\mathbf{z})$ by $\mathbf{cond}_P(\mathbf{z}) = \limsup_{\mathbf{x} \rightarrow \mathbf{z}} \mathbf{cond}_P(\mathbf{x})$.

Note that although the condition number depends on P and on the choice of norm, $\mathbf{cond}(\mathbf{z}) = \mathbf{cond}_P(\mathbf{z})$ is the condition number of the data $\mathbf{z}$ — not the condition number of the solution $P(\mathbf{z})$ and not the condition number of an algorithm that may be used to evaluate $P(\mathbf{z})$.

Facts:

For proofs and additional background, see, for example, [Dat95], [GV96], [Ste98], or [Wil65].

1. Because rounding errors are ubiquitous, a finite precision computational procedure can at best produce $P(\mathbf{z} + \delta\mathbf{z})$ where, in a suitably chosen norm, $\|\delta\mathbf{z}\| \leq \varepsilon\|\mathbf{z}\|$ and ε is a modest multiple of the unit round of the floating point system. (See section 37.6.)
2. The relative condition number determines the tight, asymptotic relative error bound

$$\frac{\|P(\mathbf{z} + \delta\mathbf{z}) - P(\mathbf{z})\|}{\|P(\mathbf{z})\|} \leq \mathbf{cond}_P(\mathbf{z}) \frac{\|\delta\mathbf{z}\|}{\|\mathbf{z}\|} + o\left(\frac{\|\delta\mathbf{z}\|}{\|\mathbf{z}\|}\right)$$

as $\delta\mathbf{z}$ tends to zero. Very roughly speaking, if the larger components of the data $\mathbf{z}$ have p correct significant digits and the condition number is $\mathbf{cond}_P(\mathbf{z}) \approx 10^s$, then the larger components of the result $P(\mathbf{z})$ have $p - s$ correct significant digits.

3. [Hig96, p. 9] If $P(\mathbf{x})$ has a Frechet derivative $D(\mathbf{z})$ at $\mathbf{z} \in \mathbb{R}^n$, then the relative condition number is

$$\mathbf{cond}_P(\mathbf{z}) = \frac{\|D(\mathbf{z})\| \|\mathbf{z}\|}{\|P(\mathbf{z})\|}.$$

In particular, if $f(x)$ is a smooth real function of a real variable x , then $\mathbf{cond}_f(z) = |zf'(z)/f(z)|$.

Examples:

1. If $P(x) = \sin(x)$ and the nominal data point $z = 22/7$ may be in error by as much as $\pi - 22/7 \approx .00126$, then $P(z) = \sin(z)$ may be in error by as much as 100%. With such an uncertainty in $z = 22/7$, $\sin(z)$ may be off by 100%, i.e., $\sin(z)$ may have relative error equal to one. In most circumstances, $z = 22/7$ is considered to be ill-conditioned.

The condition number of $z \in \mathbb{R}$ with respect to $\sin(z)$ is $\mathbf{cond}_{\sin}(z) = |z \cot(z)|$, and, in particular, $\mathbf{cond}(22/7) \approx 2485.47$. If $z = 22/7$ is perturbed to $z + \delta z = \pi$, then the asymptotic relative error bound in Fact 2 becomes

$$\begin{aligned} \left| \frac{\sin(z + \delta z) - \sin(z)}{\sin(z)} \right| &\leq \mathbf{cond}(z) \left| \frac{\delta z}{z} \right| + o(|(\delta z)/z|) \\ &= 0.9999995 \dots + o(|(\delta z)/z|). \end{aligned}$$

The actual relative error in $\sin(z)$ is $\left| \frac{\sin(z + \delta z) - \sin(z)}{\sin(z)} \right| = 1$.

2. *Subtractive Cancellation:* For $\mathbf{x} \in \mathbb{R}^2$, define $P(\mathbf{x})$ by $P(\mathbf{x}) = [1, -1]\mathbf{x}$. The gradient of $P(\mathbf{x})$ is $\nabla P(\mathbf{x}) = [1, -1]$ independent of $\mathbf{x}$, so, using the ∞ -norm, Fact 3 gives

$$\mathbf{cond}_P(\mathbf{x}) = \frac{\|\nabla f\|_{\infty} \|\mathbf{x}\|_{\infty}}{\|f(\mathbf{x})\|_{\infty}} = \frac{2 \max\{|x_1|, |x_2|\}}{|x_1 - x_2|}.$$

Reflecting the trouble associated with subtractive cancellation, $\mathbf{cond}_P(\mathbf{x})$ shows that $\mathbf{x}$ is ill-conditioned when $x_1 \approx x_2$.

3. *Conditioning of Matrix–Vector Multiplication:* More generally, for a fixed matrix $A \in \mathbb{F}^{m \times n}$ that is not subject to perturbation, define $P(\mathbf{x}) : \mathbb{F}^n \rightarrow \mathbb{F}^m$ by $P(\mathbf{x}) = A\mathbf{x}$. The relative condition number of $\mathbf{x} \in \mathbb{F}^n$ is

$$\mathbf{cond}(\mathbf{x}) = \|A\| \frac{\|\mathbf{x}\|}{\|A\mathbf{x}\|}, \quad (37.2)$$

where the matrix norm is the operator norm induced by the chosen vector norm. If A is square and nonsingular, then $\mathbf{cond}(\mathbf{x}) \leq \|A\| \|A^{-1}\|$.

4. *Conditioning of the Polynomial Zeros:* Let $q(x) = x^2 - 2x + 1$ and consider the computational task of determining the roots of $q(x)$ from the power basis coefficients $[1, -2, 1]$. Formally, the computational problem is to evaluate the function $P : \mathbb{R}^3 \rightarrow \mathbb{C}$ that maps the power basis coefficients of quadratic polynomials to their roots. If $q(x)$ is perturbed to $q(x) + \varepsilon$, then the roots change from a double root at $x = 1$ to $x = 1 \pm \sqrt{\varepsilon}$. A relative error of ε in the data $[1, -2, 1]$ induces a relative error of $\sqrt{|\varepsilon|}$ in the roots. In particular, the roots suffer an infinite rate of change at $\varepsilon = 0$. The condition number of the coefficients $[1, -2, 1]$ is infinite (with respect to root finding).

The example illustrates the fact that the problem of calculating the roots of a polynomial q from its coefficients is highly ill-conditioned when q has multiple or near multiple roots. Although it is common to say that “multiple roots are ill-conditioned,” strictly speaking, this is incorrect. It is the coefficients that are ill-conditioned because they are the initial data for the calculation.

5. [Dat95, p. 81], [Wil64, Wil65] *Wilkinson Polynomial:* Let $w(x)$ be the degree 20 polynomial

$$w(x) = (x - 1)(x - 2) \dots (x - 20) = x^{20} - 210x^{19} + 20615x^{18} \dots + 2432902008176640000.$$

The roots of $w(x)$ are the integers 1, 2, 3, $\dots$, 20. Although distinct, the roots are highly ill-conditioned functions of the power basis coefficients. For simplicity, consider only perturbations to the coefficient of x^{19} . Perturbing the coefficient of x^{19} from -210 to $-210 - 2^{-23} \approx -210 - 1.12 \times 10^{-7}$ drastically changes some of the roots. For example, the roots 16 and 17 become a complex conjugate pair approximately equal to $16.73 \pm 2.81i$.

Let $P_{16}(z)$ be the root of $\hat{w}(x) = w(x) + (z - 210)x^{19}$ nearest 16 and let $P_{17}(z)$ be the root nearest 17. So, for $z = 210$, $P_{16}(z) = 16$ and $P_{17}(z) = 17$. The condition numbers of $z = 210$ with respect to P_{16} and P_{17} are $\mathbf{cond}_{16}(210) = 210(16^{19}/(16w'(16))) \approx 3 \times 10^{10}$ and $\mathbf{cond}_{17}(210) = 210(17^{19}/(17w'(17))) \approx 2 \times 10^{10}$, respectively. The condition numbers are so large that even perturbations as small as 2^{-23} are outside the asymptotic region in which $o\left(\frac{\|\delta \mathbf{z}\|}{\|\mathbf{z}\|}\right)$ is negligible in Fact 2.

37.5 Conditioning of Linear Systems

This section applies conditioning concepts to the computational task of finding a solution to the system of linear equations $\mathbf{Ax} = \mathbf{b}$ for a given matrix $A \in \mathbb{R}^{n \times n}$ and right-hand side vector $\mathbf{b} \in \mathbb{R}^n$.

Throughout this section, $A \in \mathbb{R}^{n \times n}$ is nonsingular. Let the matrix norm $\|\cdot\|$ be an operator matrix norm induced by the vector norm $\|\cdot\|$. Use $\|A\| + \|\mathbf{b}\|$ to measure the magnitude of the data A and $\mathbf{b}$. If $E \in \mathbb{R}^{n \times n}$ is a perturbation of A and $\mathbf{r} \in \mathbb{R}^n$ is a perturbation of $\mathbf{b}$, then $\|E\| + \|\mathbf{r}\|$ is the magnitude of the perturbation to the linear system $\mathbf{Ax} = \mathbf{b}$.

Definitions:

The norm-wise **condition number** of a nonsingular matrix A (for solving a linear system) is $\kappa(A) = \|A^{-1}\| \|A\|$. If A is singular, then by convention, $\kappa(A) = \infty$. For a specific norm $\|\cdot\|_\mu$, the condition number of A is denoted $\kappa_\mu(A)$.

Facts:

For proofs and additional background, see, for example, [Dat95], [GV96], [Ste98], or [Wil65].

1. Properties of the Condition Number:

- (a) $\kappa(A) \geq 1$.
- (b) $\kappa(AB) \leq \kappa(A)\kappa(B)$.
- (c) $\kappa(\alpha A) = \kappa(A)$, for all scalars $\alpha \neq 0$.
- (d) $\kappa_2(A) = 1$ if and only if A is a nonzero scalar multiple of an orthogonal matrix, i.e., $A^T A = \alpha I$ for some scalar α .
- (e) $\kappa_2(A) = \kappa_2(A^T)$.
- (f) $\kappa_2(A^T A) = (\kappa_2(A))^2$.
- (g) $\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2 = \sigma_{\max}/\sigma_{\min}$, where $\sigma_{\max}$ and $\sigma_{\min}$ are the largest and smallest singular values of A .

2. For the p -norms (including $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_\infty$),

$$\frac{1}{\kappa(A)} = \min \left\{ \frac{\|\delta A\|}{\|A\|} \mid A + \delta A \text{ is singular} \right\}.$$

So, $\kappa(A)$ is one over the relative-to- $\|A\|$ distance from A to the nearest singular matrix, and, in particular, $\kappa(A)$ is large if and only if a small-relative-to- $\|A\|$ perturbation of A is singular.

3. Regarding A as fixed and not subject to errors, it follows from Equation 37.2 that the condition number of $\mathbf{b}$ with respect to solving $\mathbf{Ax} = \mathbf{b}$ as defined in Equation 37.1 is

$$\mathbf{cond}(\mathbf{b}) = \frac{\|A^{-1}\| \|\mathbf{b}\|}{\|A^{-1}\mathbf{b}\|} \leq \kappa(A).$$

If the matrix norm is $\|A^{-1}\|$ is induced by the vector norm $\|\mathbf{b}\|$, then equality is possible.

4. Regarding $\mathbf{b}$ as fixed and not subject to errors, the condition number of A with respect to solving $A\mathbf{x} = \mathbf{b}$ as defined in Equation 37.1 is $\mathbf{cond}(A) = \|A^{-1}\| \|A\| = \kappa(A)$.
5. $\kappa(A) \leq \mathbf{cond}([A, \mathbf{b}]) \leq \left(\frac{\|A\| + \|\mathbf{b}\|}{\|A\| \|\mathbf{b}\|} \right) \kappa(A)$, where $\mathbf{cond}([A, \mathbf{b}])$ is the condition number of the data $[A, \mathbf{b}]$ with respect to solving $A\mathbf{x} = \mathbf{b}$ as defined in Equation 37.1. Hence, the data $[A, \mathbf{b}]$ are norm-wise ill-conditioned for the problem of solving $A\mathbf{x} = \mathbf{b}$ if and only if $\kappa(A)$ is large.
6. If $\mathbf{r} = \mathbf{b} - A(\mathbf{x} + \delta\mathbf{x})$, then the 2-norm and Frobenius norm smallest perturbation $\delta A \in \mathbb{R}^{n \times n}$ satisfying $(A + \delta A)(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b}$ is $\delta A = \frac{\mathbf{r}\mathbf{x}^T}{\mathbf{x}^T\mathbf{x}}$ and $\|\delta A\|_2 = \|\delta A\|_F = \frac{\|\mathbf{r}\|_2}{\|\mathbf{x}\|_2}$.
7. Let δA and $\delta\mathbf{b}$ be perturbations of the data A and $\mathbf{b}$, respectively. If $\|A^{-1}\delta A\| < 1$, then $A + \delta A$ is nonsingular, there is a unique solution $\mathbf{x} + \delta\mathbf{x}$ to $(A + \delta A)(\mathbf{x} + \delta\mathbf{x}) = (\mathbf{b} + \delta\mathbf{b})$, and

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\|A\| \|A^{-1}\|}{(1 - \|A^{-1}\delta A\|)} \left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|} \right).$$

Examples:

1. *An Ill-Conditioned Linear System:* For $\varepsilon \in \mathbb{R}$, let $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 + \varepsilon \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. For $\varepsilon \neq 0$, A is

nonsingular and $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ satisfies $A\mathbf{x} = \mathbf{b}$. The system of equations is ill-conditioned when ε is small because some small changes in the data cause a large change in the solution. For example, perturbing $\mathbf{b}$ to $\mathbf{b} + \delta\mathbf{b}$, where $\delta\mathbf{b} = \begin{bmatrix} 0 \\ \varepsilon \end{bmatrix} \in \mathbb{R}^2$, changes the solution $\mathbf{x}$ to $\mathbf{x} + \delta\mathbf{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ independent of the choice of ε no matter how small.

Using the 1-norm, $\kappa_1(A) = \|A^{-1}\|_1 \|A\|_1 = (2 + \varepsilon)^2 \varepsilon^{-1}$. As ε tends to zero, the perturbation $\delta\mathbf{b}$ tends to zero, but the condition number $\kappa_1(A)$ explodes to infinity.

Geometrically, $\mathbf{x}$ gives the coordinates of the intersection of the two lines $x + y = 1$ and $x + (1 + \varepsilon)y = 1$. If ε is small, then these lines are nearly parallel, so a small change in them may move the intersection a long distance.

Also notice that the singular matrix $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ is a ε perturbation of A .

2. *A Well-Conditioned Linear System Problem:* Let $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. For $\mathbf{b} \in \mathbb{R}^2$, the solution to $A\mathbf{x} = \mathbf{b}$

is $\mathbf{x} = \frac{1}{2} \begin{bmatrix} b_1 + b_2 \\ b_1 - b_2 \end{bmatrix}$. In particular, perturbing $\mathbf{b}$ to $\mathbf{b} + \delta\mathbf{b}$ changes $\mathbf{x}$ to $\mathbf{x} + \delta\mathbf{x}$ with $\|\delta\mathbf{x}\|_1 \leq \|\mathbf{b}\|_1$

and $\|\delta\mathbf{x}\|_2 = \|\delta\mathbf{b}\|_2$, i.e., $\mathbf{x}$ is perturbed by no more than $\mathbf{b}$ is perturbed. This is a well-conditioned system of equations.

The 1-norm condition number of A is $\kappa_1(A) = 2$, and the 2-norm condition number is $\kappa_2(A) = 1$, which is as small as possible.

Geometrically, $\mathbf{x}$ gives the coordinates of the intersection of the perpendicular lines $x + y = 1$ and $x - y = 1$. Slightly perturbing the lines only slightly perturbs their intersection.

Also notice that for both the 1-norm and 2-norm $\min_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| = 1$, so no small-relative-to- $\|A\|$ perturbation of A is singular. If $A + \delta A$ is singular, then $\|\delta A\| \geq 1$.

3. *Some Well-known Ill-conditioned Matrices:*

(a) The upper triangular matrices $B_n \in \mathbb{R}^n$ of the form

$$B_3 = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

have ∞ -norm condition number $\kappa_\infty = n2^{n-1}$. Replacing the $(n, 1)$ entry by -2^{2-n} makes B_n singular. Note that the determinant $\det(B_n) = 1$ gives no indication of how nearly singular the matrices B_n are.

- (b) *The Hilbert matrix:* The order n Hilbert matrix $H_n \in \mathbb{R}^{n \times n}$ is defined by $h_{ij} = 1/(i + j - 1)$. The Hilbert matrix arises naturally in calculating best L_2 polynomial approximations. The following table lists the 2-norm condition numbers to the nearest power of 10 of selected Hilbert matrices.

n :	1	2	3	4	5	6	7	8	9	10
$\kappa_2(H_n)$:	1	10	10^3	10^4	10^5	10^7	10^8	10^{10}	10^{11}	10^{13}

- (c) *Vandermonde matrix:* The Vandermonde matrix corresponding to $\mathbf{x} \in \mathbb{R}^n$ is $V_{\mathbf{x}} \in \mathbb{R}^{n \times n}$ given by $v_{ij} = x_i^{n-j}$. Vandermonde matrices arise naturally in polynomial interpolation computations. The following table lists the 2-norm condition numbers to the nearest power of 10 of selected Vandermonde matrices.

n :	1	2	3	4	5	6	7	8	9	10
$\kappa_2(V_{[1,2,3,\dots,n]})$:	1	10	10	10^3	10^4	10^5	10^7	10^9	10^{10}	10^{12}

37.6 Floating Point Numbers

Most scientific and engineering computations rely on **floating point arithmetic**. At this writing, the IEEE 754 standard of binary floating point arithmetic [IEEE754] and the IEEE 854 standard of radix-independent floating point arithmetic [IEEE854] are the most widely accepted standards for floating point arithmetic. The still incomplete revised floating point arithmetic standard [IEEE754r] is planned to incorporate both [IEEE754] and [IEEE854] along with extensions, revisions, and clarifications. See [Ove01] for a textbook introduction to IEEE standard floating point arithmetic.

Even 20 years after publication of [IEEE754], implementations of floating point arithmetic vary in so many different ways that few axiomatic statements hold for all of them. Reflecting this unfortunate state of affairs, the summary of floating point arithmetic here is based upon IEEE 754r draft standard [IEEE754r] (necessarily omitting most of it), with frequent digressions to nonstandard floating point arithmetic.

In this section, the phrase *standard-conforming* refers to the October 20, 2005 IEEE 754r draft standard.

Definitions:

A p -digit, radix b **floating point number** with exponent bounds $e_{\max}$ and $e_{\min}$ is a real number of the form $x = \pm \left(\frac{m}{b^{p-1}}\right) b^e$, where e is an integer **exponent**, $e_{\min} \leq e \leq e_{\max}$, and m is a p -digit, base b integer **significand**. The related quantity m/b^p is called the **mantissa**. Virtually all floating point systems allow $m = 0$ and $b^{p-1} \leq m < b^p$. Standard-conforming, floating point systems allow all significands $0 \leq m < b^p$. If two or more different choices of significand m and exponent e yield the same floating point number, then the largest possible significand m with smallest possible exponent e is preferred.

In addition to finite floating point numbers, standard-conforming, floating point systems include elements that are not numbers, including ∞ , $-\infty$, and not-a-number elements collectively called NaNs. Invalid or indeterminate arithmetic operations like $0/0$ or $\infty - \infty$ as well as arithmetic operations involving NaNs result in NaNs.

The representation $\pm(m/b^{p-1})b^e$ of a floating point number is said to be **normalized** or **normal**, if $b^{p-1} \leq m < b^p$.

Floating point numbers of magnitude less than $b^{e_{\min}}$ are said to be **subnormal**, because they are too small to be normalized. The term **gradual underflow** refers to the use of subnormal floating point numbers. Standard-conforming, floating point arithmetic allows gradual underflow.

For $x \in \mathbb{R}$, a **rounding mode** maps x to a floating point number $\text{fl}(x)$. Except in cases of overflow discussed below, $\text{fl}(x)$ is either the smallest floating point number greater than or equal to x or the largest floating point number less than or equal to x . Standard-conforming, floating point arithmetic allows program control over which choice is used. The default rounding mode in standard conforming arithmetic is **round-to-nearest, ties-to-even** in which, except for overflow (described below), $\text{fl}(x)$ is the nearest floating point number to x . In case there are two floating point numbers equally distant from x , $\text{fl}(x)$ is the one with even significand.

Underflow occurs in $\text{fl}(x) = 0$ when $0 < |x| \leq b^{e_{\min}}$. Often, underflows are set quietly to zero. **Gradual underflow** occurs when $\text{fl}(x)$ is a subnormal floating point number. **Overflow** occurs when $|x|$ equals or exceeds a threshold at or near the largest floating point number $(b - b^{1-p})b^{e_{\max}}$. Standard-conforming arithmetic allows some, very limited program control over the overflow and underflow threshold, whether to set overflows to $\pm\infty$ and whether to trap program execution on overflow or underflow in order to take corrective action or to issue error messages. In the default round-to-nearest, ties-to-even rounding mode, overflow occurs if $|x| \geq (b - \frac{1}{2}b^{1-p})b^{e_{\max}}$, and in that case, $\text{fl}(x) = \pm\infty$ with the sign chosen to agree with the sign of x . By default, program execution continues without traps or interruption.

A variety of terms describe the precision with which a floating point system models real numbers.

- The **precision** is the number p of base- b digits in the significand.
- **Big M** is the largest integer M with the property that all integers $1, 2, 3, \dots, M$ are floating point numbers, but $M + 1$ is not a floating point number. If the exponent upper bound $e_{\max}$ is greater than the precision p , then $M = b^p$.
- The **machine epsilon**, $\epsilon = b^{1-p}$, is the distance between the number one and the next larger floating point number.
- The **unit round** $u = \inf \{\delta > 0 \mid \text{fl}(1 + \delta) > 1\}$. Depending on the rounding mode, u may be as large as the machine epsilon ϵ . In round-to-nearest, ties-to-even rounding mode, $u = \frac{1}{2}\epsilon$.

In standard-conforming, floating point arithmetic, if α and β are floating point numbers, then **floating point addition** $\oplus$, **floating point subtraction** $\ominus$, **floating point multiplication** $\otimes$, and **floating point division** $\oslash$ are defined by

$$\alpha \oplus \beta = \text{fl}(\alpha + \beta), \quad (37.3)$$

$$\alpha \ominus \beta = \text{fl}(\alpha - \beta), \quad (37.4)$$

$$\alpha \otimes \beta = \text{fl}(\alpha \times \beta), \quad (37.5)$$

$$\alpha \oslash \beta = \text{fl}(\alpha \div \beta), \quad (37.6)$$

The IEEE 754r [IEEE754r] standard also includes a fused addition-multiply operation that evaluates $\alpha\beta + \gamma$ with only one rounding error.

In particular, if the exact, infinitely precise value of $\alpha + \beta$, $\alpha - \beta$, $\alpha \times \beta$, or $\alpha \div \beta$ is also a floating point number, then the corresponding floating point arithmetic operation occurs without rounding error. Floating point sums, products, and differences of small integers have zero rounding error.

Nonstandard-conforming, floating point arithmetics do not always conform to this definition, but often they do. Even when they deviate, it is nearly always the case that if $\bullet$ is one of the arithmetic operations $+$, $-$, $\times$, or $\div$ and $\odot$ is the corresponding nonstandard floating point operation, then $\alpha \odot \beta$ is a floating point number satisfying $\alpha \odot \beta = \alpha(1 + \delta\alpha) \bullet \beta(1 + \delta\beta)$ with $|\delta\alpha| \leq b^{2-p}$ and $|\delta\beta| \leq b^{2-p}$.

If $\bullet$ is one of the arithmetic operations $+$, $-$, $\times$, or $\div$ and $\odot$ is the corresponding floating point operation, then the **rounding error** in $\alpha \odot \beta$ is $(\alpha \bullet \beta) - (\alpha \odot \beta)$, i.e., rounding error is the difference between the exact, infinitely precise arithmetic operation and the floating point arithmetic operation. In more extensive calculations, **rounding error** refers to the cumulative effect of the rounding errors in the individual floating point operations.

In machine computation, **truncation error** refers to the error made by replacing an infinite process by a finite process, e.g., truncating an infinite series of numbers to a finite partial sum.

Many computers implement floating point numbers of two or more different precisions. Typical **single precision** floating point numbers have machine epsilon roughly 10^{-7} and precision roughly 7 decimal digits or 24 binary digits. Typical **double precision** floating point numbers have machine epsilon roughly 10^{-16} and precision roughly 16 decimal digits or 53 binary digits. Specification of IEEE standard arithmetic [IEEE754r] includes these three precisions. See the table in Example 1. In addition, it is not unusual to also implement **extended precision** floating point numbers with even greater precision.

If $\hat{x} \in \mathbb{F}$ is an approximation to $x \in \mathbb{F}$, the **absolute error** in $\hat{x}$ is $|x - \hat{x}|$ and the **relative error** in $\hat{x}$ is $|(x - \hat{x})/x|$. If $x = 0$, then the relative error is undefined.

Subtractive cancellation of significant digits occurs in floating point sums when the relative error in the rounding-error-corrupted approximate sum is substantially greater than the relative error in the summands. In cases of subtractive cancellation, the sum has magnitude substantially smaller than the magnitude of the individual summands.

Facts:

For proofs and additional background, see, for example, [Ove01].

In this section, we make the following assumptions.

- The numbers α , β , and γ are p -digit radix b floating point numbers with exponent bounds $e_{\max}$ and $e_{\min}$.
- The floating point arithmetic operations satisfy Equation 37.3 to Equation 37.6.
- In the absence of overflows, the rounding mode $\text{fl}(x)$ maps $x \in \mathbb{R}$ to the closest floating point number or to one of the two closest in case of a tie. In particular, the unit round is $\frac{1}{2}b^{1-p}$.

Standard-conforming arithmetic in round-to-nearest, ties-to-even rounding mode satisfies these assumptions.

For vectors $\mathbf{x} \in \mathbb{R}^n$ and matrices $M \in \mathbb{R}^{m \times n}$, the notation $|\mathbf{x}|$ and $|M|$ indicates the vector and matrix whose entries are the absolute values of the corresponding entries of $\mathbf{x}$ and M , respectively. For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ the inequality $\mathbf{x} \leq \mathbf{y}$ represents the n scalar inequalities $x_i \leq y_i$, $i = 1, 2, 3, \dots, n$. Similarly, for $A, B \in \mathbb{R}^{m \times n}$, the inequality $A \leq B$ represents the mn inequalities $a_{ij} \leq b_{ij}$, $i = 1, 2, 3, \dots, m$ and $j = 1, 2, 3, \dots, n$.

If e is an arithmetic expression involving only floating point numbers, the notation $\text{fl}(e)$ represents the expression obtained by replacing each arithmetic operation by the corresponding floating point arithmetic operation 37.3, 37.4, 37.5, or 37.6. Note that $\text{fl}(\cdot)$ is not a function, because its value may depend on the order of operations in e —not value of e .

1. At this writing, the only radices in common use on computers and calculators are $b = 2$ and $b = 10$.
2. The commutative properties of addition and multiplication hold for floating point addition 37.3 and floating point multiplication 37.5, i.e., $\alpha \oplus \beta = \beta \oplus \alpha$ and $\alpha \otimes \beta = \beta \otimes \alpha$. (Examples below show that the associative and distributive properties of addition and multiplication *do not* in general hold for floating point arithmetic.)
3. In the absence of overflow or any kind of underflow, $\text{fl}(x) = x(1 + \delta)$ with $|\delta| < \frac{1}{2}b^{1-p}$.
4. *Rounding in Arithmetic Operations:* If $\bullet$ is an arithmetic operation and $\odot$ is the corresponding floating point operation, then

$$\alpha \odot \beta = (\alpha \bullet \beta)(1 + \delta)$$

with $|\delta| \leq \frac{1}{2}b^{1-p}$. The error δ may depend on α and β as well as the arithmetic operation.

5. *Differences of Nearby Floating Point Numbers:* If $\alpha > 0$ and $\beta > 0$ are normalized floating point numbers and $\frac{1}{2} < \alpha/\beta < 2$, then $\text{fl}(\alpha - \beta) = \alpha \ominus \beta = \alpha - \beta$, i.e., there is zero rounding error in floating point subtraction of nearby numbers.